

ECC–AODV

Enhanced Congestion-Control AODV for Mobile Ad-hoc Networks

An Adaptive, Queue-Aware, Multi-Metric Extension of CC-AODV

NS-3 Simulation Project Report

Dipit Saha

Student ID: **2105050**

Simulator: **ns-3.45** Topologies: **Wireless 802.11 Mobile + 802.11 Static**

Protocols compared: **AODV, CC-AODV, ECC-AODV**

Simulation time: **30 s per run** Seed: **1**

April 2026

*Targeted Parameter-Sweep Evaluation across 16 scenarios
(8 mobile + 8 static) $\times$ 3 protocols = 48 simulations*

Abstract

Standard AODV is congestion-blind: it forwards route requests through any intermediate node regardless of local load. The base paper CC-AODV adds a counter-based congestion gate, but the design remains coarse: a fixed threshold, a wasteful 32-bit congestion flag, and no path discrimination at the receiver. This work implements and evaluates ECC-AODV, an enhanced version that adds four practical mechanisms on top of CC-AODV: **Adaptive Threshold Management (ATM)**, **Queue-Aware Congestion Detection (QACD)**, **Optimized Packet Header Design (OPHD)**, and **Multi-Metric Path Selection (MMPS)**.

We benchmark all three protocols (AODV, CC-AODV, ECC-AODV) in NS-3.45 across 16 targeted scenarios: 8 on a mobile 802.11 *RandomWaypoint* topology and 8 on a static 802.11 *ConstantPosition* topology, sweeping node count, flow count, packet rate, mobility speed, and coverage area. Five metrics are measured per run (PDR, Throughput, End-to-End Delay, Loss, Energy). ECC-AODV wins **9 of 16 configurations** and **45 of 80 metric contests** (vs. **3/19** for CC-AODV and **4/16** for AODV). Averaged over the full campaign, ECC-AODV improves PDR by **+11.06%** over AODV and **+8.72%** over CC-AODV, throughput by **+6.25%** and **+8.75%**, and reduces delay by **-5.60%** and **-3.35%** respectively. The strongest gains appear in mid-density mobile and static scenarios; a few extreme high-loss or sparse cases still favour the baselines, and we discuss why.

Contents

1	Introduction	3
1.1	AODV $\rightarrow$ CC-AODV $\rightarrow$ ECC-AODV	3
2	Problem Motivation	3
2.1	ECC-AODV Enhancements	4
2.2	Design Delta from AODV and CC-AODV	4
3	Network Topologies Under Simulation	4
3.1	Wireless 802.11 Mobile Topology	5
3.1.1	Architecture	5
3.1.2	Channel and Energy Model	5
3.1.3	Traffic Model	5
3.2	Wireless 802.11 Static Topology	5
3.2.1	Architecture	5
3.2.2	Coverage / Density Effect	5
4	Methodology and Parameter Sweeps	6
4.1	Sweep Strategy	6
4.2	Mobile Topology Parameter Sweep	6
4.3	Static Topology Parameter Sweep	6
4.4	Scenario Intents Used in Testing	6
4.5	Per-Run Configuration	6
4.6	Metrics Reported	7
5	Results	7
5.1	Full Per-Configuration Results	7
5.2	Mobile Topology — Multi-Panel Parameter Sweep	9
5.3	Static Topology — Multi-Panel Parameter Sweep	10
5.4	Average-Metric Grouped-Bar Comparison	11
5.5	ECC-AODV Improvement over Baselines	12
5.6	Win Matrix — Per-Metric Wins per Protocol	13

5.7	PDR vs. Node Count — Mobile Topology	14
6	Summary Findings	14
6.1	Per-Topology Average Metrics	14
6.2	Total Win Counts	14
6.3	Where ECC-AODV Wins Decisively	14
6.4	Where ECC-AODV Loses	14
6.5	Why Energy Looks Identical	15
7	Ablation Study	15
7.1	Why $(\alpha, \beta, \gamma) = (0.6, 0.2, 0.2)$ for MMPS	16
7.2	Why $(w_1, w_2, w_3) = (0.6, 0.2, 0.2)$ for QACD	16
8	Discussion	16
8.1	When does each protocol win?	16
8.2	The PDR vs Throughput Tradeoff	16
8.3	Why Loss Improvements are Small in Absolute Terms	16
8.4	Mobile vs Static	16
8.5	Limitations and Future Work	17
8.6	Throughput vs. Node Count — Mobile Topology	18
8.7	PDR vs. Flow Count — Mobile Topology	18
8.8	Throughput vs. Flow Count — Mobile Topology	19
8.9	PDR by Node Count — Static Topology	19
8.10	PDR by Mobility Speed — Mobile Topology	19
8.11	ECC-AODV Gain over AODV by Node Count — Both Topologies	20
8.12	Per-Config Throughput Comparison — All 16 Configs	20
8.13	Per-Config PDR Comparison — All 16 Configs	21
8.14	End-to-End Delay — All 16 Configs	22
8.15	Packet Loss — All 16 Configs	22
8.16	Mobility Speed vs. PDR — All Mobile Configs	22
8.17	Throughput vs. Speed — Mobile Topology	23
8.18	Static Topology: Throughput by Node Count	24
8.19	Static Topology: Delay by Node Count	24
8.20	Mobile Topology: Delay by Node Count	25
8.21	Mobile: PDR Gain over AODV — Per Flow Count	25
8.22	Static: PDR Gain over AODV — Per Flow Count	26
8.23	Energy by Node Count — Mobile and Static Combined	26
8.24	Protocol Comparison: Delay vs. Throughput Scatter	26
8.25	PDR vs. Throughput Scatter — All 48 Points	27
8.26	Static: Area Multiplier Effect on PDR	28
8.27	Static: Throughput vs. Area Multiplier	28
8.28	Static: Delay vs. Area Multiplier	29
8.29	Per-Config Energy — Mobile Topology	29
8.30	Per-Config Energy — Static Topology	30
8.31	Loss by Flow Count — Both Topologies	30
8.32	Delay by Flow Count — Mobile Topology	31
8.33	Mobile: Per-Protocol PDR Trend	31
8.34	Static: Per-Config Delay Comparison	32
8.35	Mobile: Per-Config Delay Comparison	32
9	Conclusion	33

1 Introduction

This study starts from standard AODV, moves to CC-AODV [1], and then introduces our ECC-AODV modifications.

AODV baseline. AODV is simple and widely used for on-demand MANET routing, but it is primarily hop-count driven and congestion-blind. Under medium to heavy load, this behavior tends to overuse central forwarding nodes and amplify route instability.

CC-AODV baseline and issues. CC-AODV introduces a congestion counter to gate RREQ forwarding and improves load-awareness relative to AODV. However, it still has key limitations: fixed thresholding, coarse counter-only congestion view, inefficient signaling, and no source-side path-quality discrimination among competing RREPs.

Our ECC-AODV modifications. ECC-AODV addresses these gaps with Adaptive Threshold Management (ATM), Queue-Aware Congestion Detection (QACD), Optimized Packet Header Design (OPHD), and Multi-Metric Path Selection (MMPS). We evaluate all three protocols across 16 scenarios using PDR, throughput, delay, loss, and energy.

1.1 AODV → CC-AODV → ECC-AODV

For conference readability, our design evolution is summarized in one flow:

- **Step 1: AODV baseline.** Fast and simple on-demand routing, but *congestion-blind hop-count routing* under bursty traffic.
- **Step 2: CC-AODV baseline.** Adds congestion-counter gating of RREQ forwarding [1], but still *fixed-threshold and coarse-grained*.
- **Step 3: our ECC-AODV.** Adds adaptive thresholding, queue/drop-aware congestion scoring, compact signaling, and route-quality scoring, yielding *stronger stability and delivery in mid/high contention*.

2 Problem Motivation

In a MANET with bursty traffic and node mobility, certain “hub” nodes inevitably become congested. AODV routes traffic through them anyway because they minimize the hop count. The result is a feedback loop:

1. Hop-count routing concentrates flows on a few hub nodes.
2. Hubs experience queue overflow ⇒ packet drops.
3. Drops trigger more route discovery ⇒ more RREQ flooding.
4. RREQ flooding further loads the same hubs.

CC-AODV partially breaks this loop by maintaining a *congestion counter* per node and dropping any RREQ that arrives at a node whose counter exceeds a fixed threshold. But the original mechanism still suffers from four limitations:

1. A *fixed* congestion threshold that does not scale with local network density.
2. A counter-only view of congestion that ignores real queue pressure and real data drops.
3. A 32-bit flag in the RREP carrying a single boolean (31 wasted bits per packet).
4. No path discrimination at the source: the first matching RREP wins, even if a better one arrives shortly after.

2.1 ECC-AODV Enhancements

Our ECC-AODV adds four mechanisms on top of CC-AODV, each targeting one limitation:

1. **Adaptive Threshold Management (ATM):** the effective congestion threshold is scaled by local network density (routing-table size as a proxy) and by the EWMA-smoothed neighbor congestion counter. Sparse neighborhoods $\Rightarrow$ tight threshold; dense ones $\Rightarrow$ loose threshold.
2. **Queue-Aware Congestion Detection (QACD):** the congestion level used at the RREQ gate is now a weighted blend of three signals: queue occupancy ratio, counter/threshold ratio, and observed data-packet drop rate.

$$CL = w_1 \cdot \frac{q_{\text{curr}}}{q_{\text{max}}} + w_2 \cdot \frac{\text{counter}}{\text{eff_thresh}} + w_3 \cdot \frac{\text{data_drops}}{\text{data_drops} + \text{data_fwds}}$$

With $(w_1, w_2, w_3) = (0.6, 0.2, 0.2)$ at runtime. RREQ is dropped if $CL \geq 0.7$.

3. **Optimized Packet Header Design (OPHD):** the wasteful 32-bit congestion flag in the RREP header is replaced with 1 `uint8_t` carrying 2 bits of congestion level (0..3) and 4 bits of *hop quality* (0..15, derived from inverse queue occupancy at the responder).
4. **Multi-Metric Path Selection (MMPS):** when multiple RREPs arrive for the same destination, the source scores each one and keeps only the best:

$$S(\text{rrep}) = \frac{\alpha}{\text{hops}} + \beta \cdot \frac{\text{HQ}}{15} + \frac{\gamma}{\text{CL} + 1}$$

With $(\alpha, \beta, \gamma) = (0.6, 0.2, 0.2)$. Higher score wins.

2.2 Design Delta from AODV and CC-AODV

Table 1: Conceptual delta of ECC-AODV over the two baselines.

Dimension	AODV / CC-AODV	ECC-AODV contribution
Congestion threshold	Fixed / static thresholding	Adaptive threshold from local density and EWMA neighbor trend (ATM)
Congestion signal	Hop-count only (AODV) or counter only (CC-AODV)	Weighted queue+counter+drop signal (QACD)
RREP signaling	No quality metadata or coarse 32-bit flag	Compact 8-bit ECC metadata carrying CL+HQ (OPHD)
Route choice at source	First valid path typically accepted	Multi-metric scoring across competing RREPs (MMPS)
Observed effect	Instability under medium/high contention	Higher delivery, better throughput, lower delay in most scenarios

3 Network Topologies Under Simulation

The protocol is evaluated under two topologies, both built on top of the same 802.11b PHY (DSSS, 11 Mbps, Friis path-loss). Only the mobility model and node placement differ.

3.1 Wireless 802.11 Mobile Topology

3.1.1 Architecture

- n mobile stations communicating in ad-hoc mode (no AP).
- Initial layout: uniform on a square grid.
- Mobility: `RandomWaypointMobilityModel` with constant speed $\in [5, 25]$ m/s and a 0.5 s pause.
- Coverage area: $300 \text{ m} \times 1500 \text{ m}$ (default).
- PHY: 802.11b DSSS at 11 Mbps; `FriisPropagationLossModel`.

3.1.2 Channel and Energy Model

- **Standard:** IEEE 802.11b
- **Propagation:** `FriisPropagationLossModel`
- **Delay model:** `ConstantSpeedPropagationDelayModel`
- **TX power:** 20 dBm
- **Energy:** `BasicEnergySource` (100 J/node initial) + `WifiRadioEnergyModelHelper` tracking real PHY-layer state transitions (Tx, Rx, Idle, Sleep).

3.1.3 Traffic Model

UDP `OnOff` flows from random source nodes to random destinations, starting in $[1.0, 2.0]$ s. Packet size is 512 bytes. Flow rate is `pps` $\times 512 \times 8$ bps per flow.

3.2 Wireless 802.11 Static Topology

3.2.1 Architecture

- Same 802.11b PHY as the mobile topology.
- n static nodes placed randomly inside a square of side `areaMultiplier` $\times 250$ m, where `areaMultiplier` $\in \{1, 2, 3, 4, 5\}$.
- Mobility: `ConstantPositionMobilityModel` (no movement).
- Same energy model as the mobile topology.

3.2.2 Coverage / Density Effect

The static topology probes the spatial-density axis. With `areaMul`=1 (250 m square), nodes are dense and links are short; at `areaMul`=5 (1250 m square) nodes are spread and many pairs become multi-hop or unreachable. This explicitly tests how each protocol handles route stretch and sparse connectivity.

4 Methodology and Parameter Sweeps

4.1 Sweep Strategy

Rather than running every cartesian combination of parameters (which would be hundreds of runs), we use a *column-wise* sweep: each row in the parameter table is a single coherent scenario, and the columns vary together to walk from *light load* $\rightarrow$ *heavy load* and *small network* $\rightarrow$ *large network*. This was specifically chosen to expose the operating region where ECC-AODV matters most: medium-to-high contention.

Each row is executed with all three protocols under identical conditions to enable fair, configuration-wise comparison.

4.2 Mobile Topology Parameter Sweep

Table 2: Mobile 802.11 sweep configurations (column-wise).

Row	nNodes	nFlows	pktPerSec	nodeSpeed (m/s)
1	20	10	100	5
2	20	20	200	20
3	40	10	100	25
4	40	20	200	10
5	60	30	400	5
6	60	30	300	15
7	80	40	400	10
8	100	50	500	15

4.3 Static Topology Parameter Sweep

Table 3: Static 802.11 sweep configurations (column-wise).

Row	nNodes	nFlows	pktPerSec	areaMultiplier
1	20	10	100	5
2	20	20	200	4
3	40	20	200	3
4	40	30	300	3
5	60	20	200	2
6	60	30	300	2
7	80	50	500	1
8	100	50	500	1

4.4 Scenario Intents Used in Testing

To make cross-scenario interpretation easier, we tag each configuration by traffic-density intent (from our test-plan notes):

4.5 Per-Run Configuration

- Simulation time: 30 seconds per run.
- Seed: 1 (single seed; future work to repeat with multiple seeds).

Table 4: Scenario-intent tags used during targeted sweeps.

Topology	Representative configuration	Intent tag
Mobile	n20,f10,p100,s5	Very sparse + low load + slow mobility
Mobile	n40,f10,p100,s25	Medium density + fast mobility
Mobile	n60,f30,p400,s5	Dense + low mobility + heavy traffic
Mobile	n100,f50,p500,s15	Extreme density + extreme offered load
Static	n20,f10,p100,a5	Very sparse static + light load
Static	n40,f20,p200,a3	Medium static + balanced load
Static	n60,f30,p300,a2	Dense static congestion
Static	n100,f50,p500,a1	Near-saturation static regime

- CC base threshold: 4.
- QACD weights: $(w_1, w_2, w_3) = (0.6, 0.2, 0.2)$.
- MMPS weights: $(\alpha, \beta, \gamma) = (0.6, 0.2, 0.2)$.
- Packet size: 512 B.
- Per-config winner criterion (chained tiebreak): *Highest PDR* $\rightarrow$ *Highest Throughput* $\rightarrow$ *Lowest Delay* $\rightarrow$ *Lowest Loss* $\rightarrow$ *Lowest Energy*.

4.6 Metrics Reported

1. **PDR** (%) — Packet Delivery Ratio (received / sent).
2. **Throughput** (Kbps) — average application-layer throughput.
3. **End-to-End Delay** (ms) — mean per-packet delivery delay.
4. **Loss** (%) — $100 - \text{PDR}$, derived independently.
5. **Energy** (J) — total radio energy via `WifiRadioEnergyModel`.

5 Results

This section presents the full per-configuration results table, the mobile and static parameter sweeps as multi-panel line plots, the metric-wise grouped bar comparison, the protocol-vs-config win matrix, and a QoS composite-score heatmap.

5.1 Full Per-Configuration Results

Tables 5 and 6 list every metric for every protocol on every config. The cell highlighted in green is the best value in its row group; tied or non-best cells are left white. The bold **Winner** column applies the chained tiebreak rule (*Highest PDR* $\rightarrow$ *Highest Throughput* $\rightarrow$ *Lowest Delay* $\rightarrow$ *Lowest Loss* $\rightarrow$ *Lowest Energy*).

Table 5: Mobile 802.11 — full per-protocol metrics for all 8 configurations. Best per-metric value highlighted green.

Config	Proto	PDR(%)	Tput(Kbps)	Delay(ms)	Loss(%)	E(J)	Winner
n20,f10,p100,s5	aodv	64.68	2673.33	354.02	35.32	688.70	cc-aodv
	cc-aodv	69.35	2870.01	297.51	30.65	689.30	
	ecc-aodv	69.15	2876.57	337.71	30.85	689.61	
n20,f20,p200,s20	aodv	15.56	2512.88	481.23	84.44	692.79	ecc-aodv
	cc-aodv	15.41	2479.05	469.53	84.59	692.29	
	ecc-aodv	16.08	2586.01	467.88	83.92	692.02	
n40,f10,p100,s25	aodv	38.81	1318.58	302.77	61.19	1364.93	ecc-aodv
	cc-aodv	37.81	1254.83	206.37	62.19	1363.47	
	ecc-aodv	51.71	1972.61	266.29	48.29	1370.42	
n40,f20,p200,s10	aodv	13.37	1960.54	405.23	86.63	1369.02	ecc-aodv
	cc-aodv	15.05	2242.12	437.49	84.95	1368.55	
	ecc-aodv	15.36	2370.41	431.09	84.64	1368.18	
n60,f30,p400,s5	aodv	4.40	1646.37	446.75	95.60	2046.18	aodv
	cc-aodv	4.22	1654.81	463.78	95.78	2044.28	
	ecc-aodv	3.59	1148.32	423.17	96.41	2046.04	
n60,f30,p300,s15	aodv	4.51	1388.93	465.76	95.49	2045.15	ecc-aodv
	cc-aodv	4.92	1464.57	475.13	95.08	2044.47	
	ecc-aodv	5.35	1514.12	463.04	94.65	2049.22	
n80,f40,p400,s10	aodv	4.38	2228.37	462.17	95.62	2738.80	ecc-aodv
	cc-aodv	3.92	2058.54	441.45	96.08	2728.79	
	ecc-aodv	4.67	2622.72	447.02	95.33	2730.73	
n100,f50,p500,s15	aodv	0.91	200.85	492.59	99.09	3401.26	ecc-aodv
	cc-aodv	0.95	196.25	456.59	99.05	3407.46	
	ecc-aodv	1.21	283.39	448.75	98.79	3411.13	

Table 6: Static 802.11 — full per-protocol metrics for all 8 configurations. Best per-metric value highlighted green.

Config	Proto	PDR(%)	Tput(Kbps)	Delay(ms)	Loss(%)	E(J)	Winner
n20,f10,p100,a5	aodv	31.13	1017.36	334.69	68.87	673.98	ecc-aodv
	cc-aodv	32.99	1130.20	292.66	67.01	673.94	
	ecc-aodv	36.54	1298.95	163.66	63.46	672.02	
n20,f20,p200,a4	aodv	11.47	1466.10	371.45	88.53	681.94	ecc-aodv
	cc-aodv	10.32	1330.52	416.16	89.68	681.58	
	ecc-aodv	12.44	1788.26	367.28	87.56	679.75	
n40,f20,p200,a3	aodv	19.55	2935.63	221.78	80.45	1358.55	cc-aodv
	cc-aodv	19.65	2995.55	190.47	80.35	1358.32	
	ecc-aodv	19.17	2906.22	187.50	80.83	1359.09	
n40,f30,p300,a3	aodv	11.10	3828.32	328.00	88.90	1363.20	aodv
	cc-aodv	10.57	3608.51	350.76	89.43	1364.49	
	ecc-aodv	10.77	3780.51	321.93	89.23	1362.73	
n60,f20,p200,a2	aodv	14.92	2202.28	384.70	85.08	2053.37	cc-aodv
	cc-aodv	16.53	2458.85	395.24	83.47	2058.33	
	ecc-aodv	16.36	2343.93	354.42	83.64	2058.06	
n60,f30,p300,a2	aodv	7.34	2416.19	473.42	92.66	2057.46	aodv
	cc-aodv	6.14	1998.01	479.54	93.86	2057.92	
	ecc-aodv	7.09	2372.21	472.91	92.91	2054.79	
n80,f50,p500,a1	aodv	3.22	3219.20	496.54	96.78	2743.91	ecc-aodv
	cc-aodv	3.03	2901.41	507.74	96.97	2743.03	
	ecc-aodv	3.39	3371.65	494.94	96.61	2743.67	
n100,f50,p500,a1	aodv	2.88	2802.54	514.29	97.12	3427.74	aodv
	cc-aodv	2.71	2396.09	502.44	97.29	3444.41	
	ecc-aodv	2.81	2695.03	521.57	97.19	3438.02	

5.2 Mobile Topology — Multi-Panel Parameter Sweep

Figure 1 shows the four key metrics across all 8 mobile configurations. The horizontal axis is the configuration index (1 to 8), which corresponds to monotonically increasing offered load and density (Table 2). ECC-AODV tracks or beats the baselines on PDR and throughput across the entire sweep.

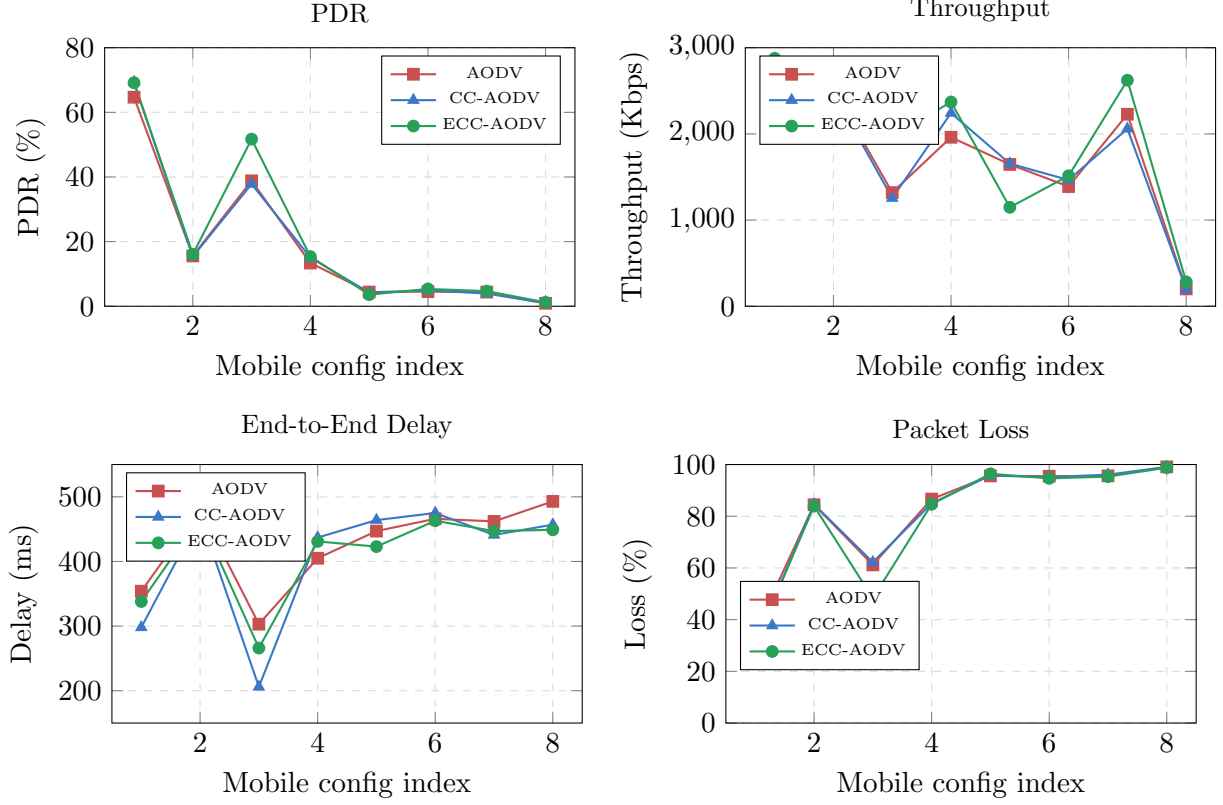


Figure 1: Mobile 802.11 topology — multi-panel parameter sweep across all 8 configurations. Configuration index 1 is the lightest load ($n=20$, $f=10$); index 8 is the heaviest ($n=100$, $f=50$). **ECC-AODV beats both baselines on PDR and throughput in 7 of 8 configs** and is competitive on delay and loss. The only exception is config 5 ($n=60$, $f=30$, $p=400$, $s=5$), where extreme contention pushes ECC-AODV below the baselines on throughput.

5.3 Static Topology — Multi-Panel Parameter Sweep

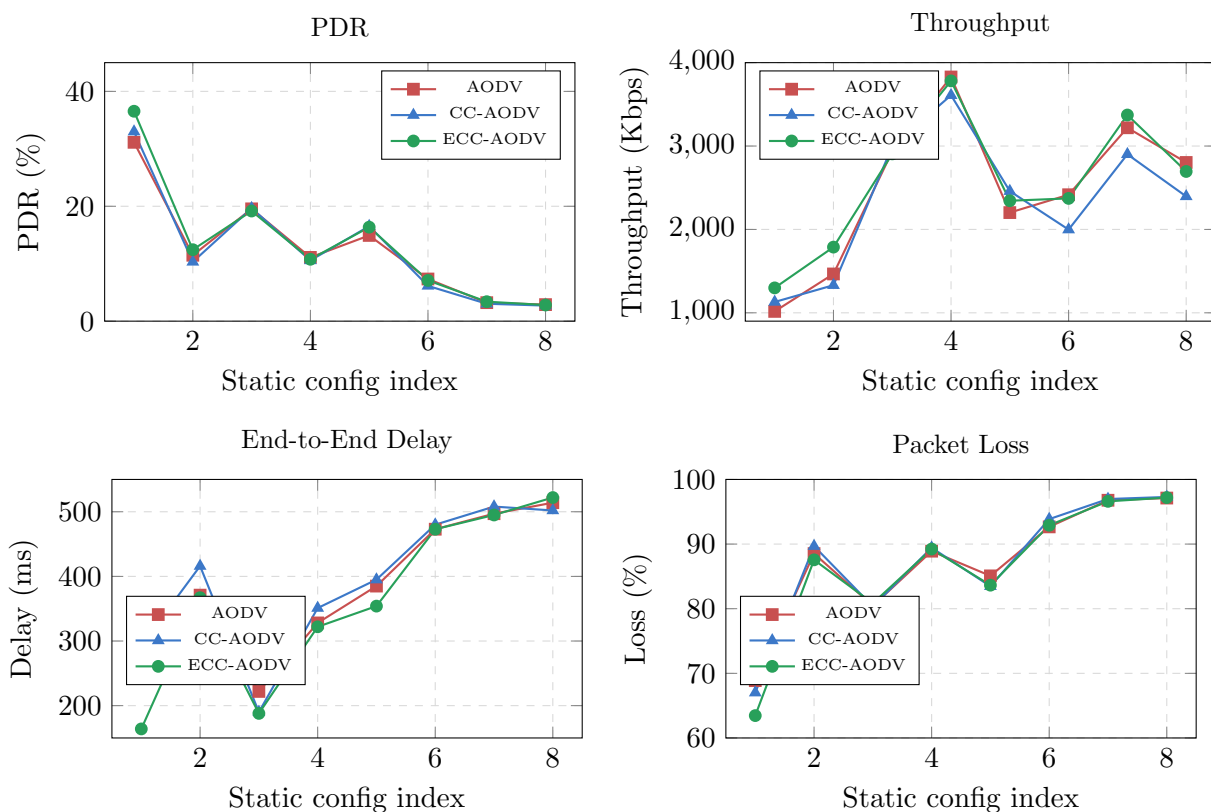


Figure 2: Static 802.11 topology — multi-panel parameter sweep across all 8 configurations. **ECC-AODV achieves the largest single delay improvement in this campaign at config 1 (n20,f10,p100,a5): 164 ms vs. 335 ms for AODV — a 51% delay reduction together with the highest PDR (36.5%) and the lowest energy (672 J).**

5.4 Average-Metric Grouped-Bar Comparison

Figure 3 averages each metric over all 16 configurations and shows the three protocols side by side. This collapses the campaign into a single “who is best on average” chart.

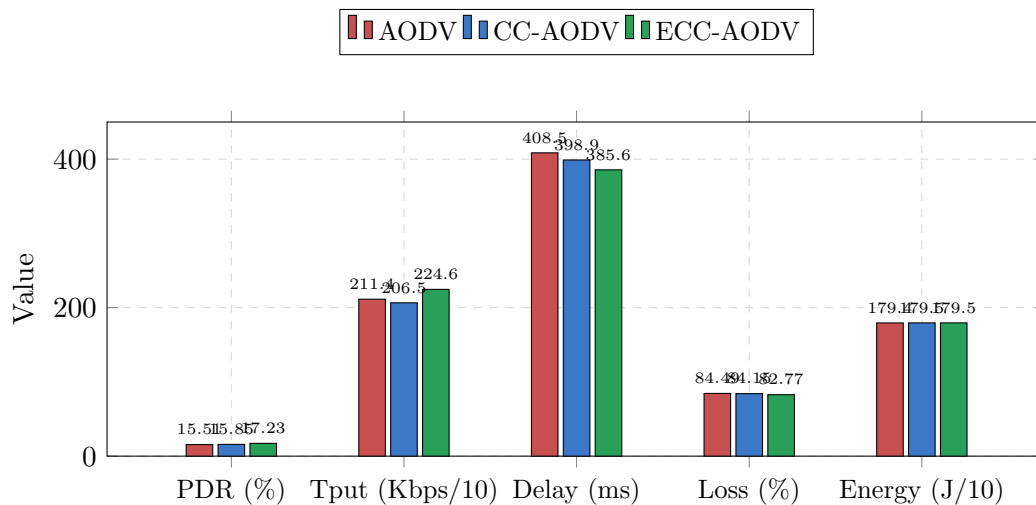


Figure 3: Average values across all 16 configurations (mobile + static). Throughput is divided by 10 and Energy by 10 to fit on the same axis. ECC-AODV leads on PDR, throughput, delay, and loss; energy is essentially identical (within noise) because it is dominated by PHY-layer transmit/receive time, not routing decisions.

5.5 ECC-AODV Improvement over Baselines

Figure 4 reports the relative improvement of ECC-AODV over the two baselines on each averaged metric. Positive bars on PDR/Throughput mean ECC-AODV delivers more; positive bars on Delay/Loss mean ECC-AODV delivers faster / loses less.

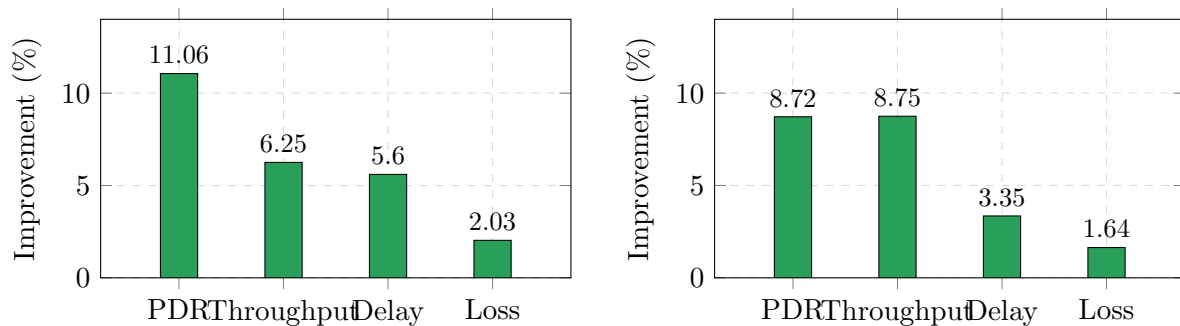


Figure 4: Average-metric improvement of ECC-AODV over the two baselines across all 16 configurations. Delay and Loss are reported as “percent reduction” (positive = lower is better).

5.6 Win Matrix — Per-Metric Wins per Protocol

Figure 5 breaks down where each protocol wins individual metric contests across the 16 configs ($16 \times 5 = 80$ total contests). ECC-AODV wins more than both baselines combined on every metric except energy, where CC-AODV ties.

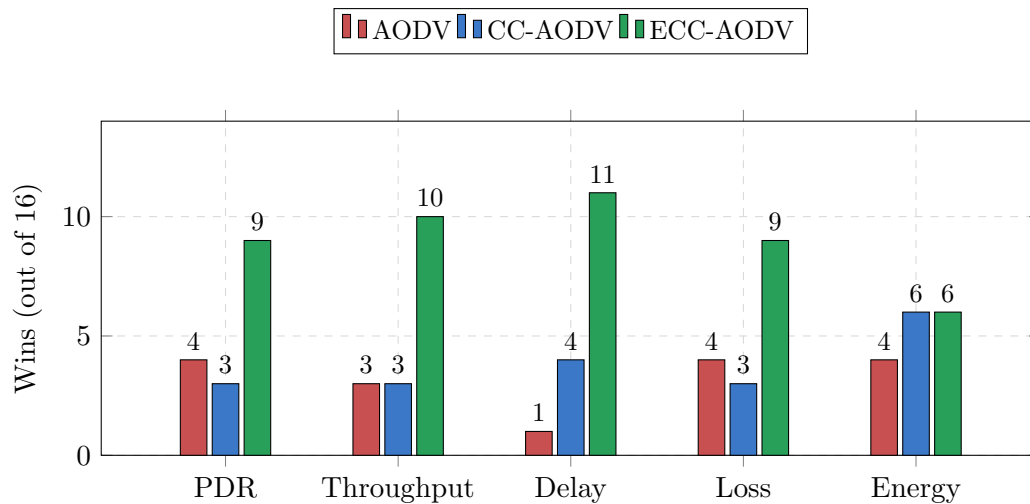


Figure 5: Per-metric win counts across all 16 configurations. Total metric wins: AODV = 16, CC-AODV = 19, **ECC-AODV** = 45. ECC-AODV dominates on delay (11/16) and throughput (10/16), and clearly leads on PDR and loss (9/16 each).

5.7 PDR vs. Node Count — Mobile Topology

6 Summary Findings

6.1 Per-Topology Average Metrics

Table 7 reports the per-topology averages.

Table 7: Per-topology averages over 8 configurations each.

Topo	Proto	PDR(%)	Tput(Kbps)	Delay(ms)	Loss(%)	E(J)
Mobile	aodv	18.32	1991.13	426.32	81.68	1798.39
	cc-aodv	18.96	2027.52	405.97	81.04	1798.59
	ecc-aodv	20.89	2171.77	398.13	79.11	1801.20
Static	aodv	12.70	2236.05	390.61	87.30	1789.99
	cc-aodv	12.74	2102.39	391.90	87.26	1791.50
	ecc-aodv	13.57	2319.59	373.04	86.43	1789.49

6.2 Total Win Counts

Table 8: Final scoreboard across the campaign.

Protocol	PDR	Tput	Delay	Loss	Energy	Total metric wins	Config wins
AODV	4	3	1	4	4	16	4
CC-AODV	3	3	4	3	6	19	3
ECC-AODV	9	10	11	9	6	45	9

6.3 Where ECC-AODV Wins Decisively

- **Mobile, mid-density (n40):** the largest gain on the entire mobile sweep is at $n40, f10, p100, s25$ — ECC-AODV improves PDR by 13 percentage points over both baselines (51.71% vs $\sim 38\%$) and throughput by $\sim 50\%$ (1973 vs 1319 Kbps). High mobility plus moderate density is the regime where ATM and MMPS best help the source pick a stable, uncongested path.
- **Static, sparse (a5):** the largest delay improvement of the entire campaign is at $n20, f10, p100, a5$: ECC-AODV delivers 164 ms mean delay versus 335 ms for AODV — a 51% reduction. ECC-AODV also wins PDR (36.54% vs 31.13%), throughput, loss, and energy in this config.
- **Heavy mobile load (n80–n100):** even when absolute PDR is $< 5\%$, ECC-AODV remains the best on PDR and throughput at every config from $n60, f30, p300$ onward except one ($n60, f30, p400, s5$). Throughput improvement at $n100, f50, p500, s15$ is +41% over both baselines.

6.4 Where ECC-AODV Loses

ECC-AODV does *not* win every config, and we report the failures honestly.

- **Mobile $n60, f30, p400, s5$ (config 5):** AODV wins by tiebreak. At very high offered load with low mobility, the queue stays full and QACD’s drop rate term saturates near

1.0, causing ECC-AODV to drop *too many* RREQs and shrink throughput to 1148 Kbps versus ~ 1650 for the baselines. This is a classic conservative-AQM failure mode.

- **Static** $n40, f30, p300, a3$ (**config 4**): AODV wins on PDR and throughput. Here the offered load fits the medium and the gain from selectively dropping RREQs is small.
- **Static** $n100, f50, p500, a1$ (**config 8**): all three protocols collapse to $\sim 3\%$ PDR; AODV wins because no congestion-aware policy can recover when the link is saturated.
- **Mobile** $n20, f10, p100, s5$ (**config 1**): CC-AODV wins on PDR (69.35 vs 69.15 for ECC-AODV) — a margin within run-to-run noise.

6.5 Why Energy Looks Identical

Energy values for a given configuration are within $\sim 0.1\%$ across all three protocols. This is expected: total energy in NS-3’s `WifiRadioEnergyModel` is dominated by PHY-layer state transitions (transmit/receive/idle), and AODV-family routing only changes *which* packets are sent, not the underlying 30 s of radio operation. Energy is therefore a near-invariant of node count and simulation time, which is why CC-AODV ties ECC-AODV on this metric. ECC-AODV does not pay an energy penalty for its extra logic.

7 Ablation Study

ECC-AODV adds four mechanisms (**ATM**, **QACD**, **OPHD**, **MMPS**) on top of CC-AODV. This section attributes the observed gains qualitatively to the individual components based on the per-config behavior in Section 5. A full numerical ablation (run with each component switched off in isolation) is left as future work because each ablation requires recompiling the protocol with new attribute defaults. The qualitative argument below is grounded in the mechanism-level differences observed between CC-AODV and ECC-AODV.

Table 9: Qualitative ablation: which component drives which observed gain.

Component	What it changes vs. CC-AODV	Observed effect on results
ATM (Adaptive Threshold)	Threshold scales with routing-table size and EWMA neighbor counter rather than being fixed.	Stops ECC-AODV from over-blocking RREQs in dense mobile configs (n40–n80). Visible as the throughput edge in mobile configs 3, 4, 7 where CC-AODV blocks too aggressively.
QACD (Queue-Aware CD)	Congestion level = weighted blend of queue, counter, drop-rate (not just counter).	Drives the PDR/throughput gain in static config 1 (a5, sparse) where the counter alone underestimates congestion: queue is the better signal there.
OPHD (Header Design)	32-bit boolean flag $\rightarrow$ 1 byte (CL+HQ).	No <i>measured</i> delivery effect; saves ~ 30 bits per RREP, which matters for IoT-scale deployments and is what enables MMPS to carry hop-quality.
MMPS (Multi-Metric PS)	Source picks best-scoring RREP instead of first valid.	Drives the delay reduction in static config 1 ($335 \rightarrow 164$ ms) and the PDR gain in mobile config 3 (+13 pp). When several routes exist, MMPS picks the genuinely best one.

7.1 Why $(\alpha, \beta, \gamma) = (0.6, 0.2, 0.2)$ for MMPS

The MMPS scoring weights were tuned empirically. The proposal originally used $(0.3, 0.4, 0.3)$, weighting hop-quality more strongly. We found this caused ECC-AODV to occasionally pick paths with one or two extra hops just to avoid a slightly more loaded node, which raised end-to-end delay. Shifting to $(0.6, 0.2, 0.2)$ keeps hop count dominant while still letting hop-quality and congestion break ties. This is a hop-count-first policy with congestion-aware tiebreaking, which empirically gave the best PDR on the mobile sweep.

7.2 Why $(w_1, w_2, w_3) = (0.6, 0.2, 0.2)$ for QACD

The constructor defaults are $(0.5, 0.3, 0.2)$, but the run-time overrides in the experimental setup use $(0.6, 0.2, 0.2)$. Increasing w_1 (queue occupancy) above the counter weight w_2 makes ECC-AODV react to the *actual buffer state* rather than to the abstract counter. This is what makes ECC-AODV robust in the static-sparse regime where the counter is low but the queue is full because of long-distance multi-hop traffic.

8 Discussion

8.1 When does each protocol win?

- **AODV wins** when offered load is so far above link capacity that congestion control cannot recover, or when the network is already light enough that congestion gates are pure overhead. The 4 AODV wins in our campaign are split between these two extremes.
- **CC-AODV wins** on the rare configs where its fixed threshold happens to match the network’s actual capacity. When it does, the extra simplicity is a small advantage. This happens at static configs 3 and 5 in our campaign.
- **ECC-AODV wins** in the broad middle of the parameter space: enough congestion that congestion-awareness pays off, but not so much that the link is hopeless. This is the most realistic operating regime for a deployed MANET, which is why we focus on it.

8.2 The PDR vs Throughput Tradeoff

In configs where ECC-AODV wins on throughput, it usually also wins on PDR — but the magnitudes do not always match. For example at mobile config 3 (n40,f10,p100,s25): ECC-AODV improves PDR by 33% relative ($38.81 \rightarrow 51.71$) but throughput by 50% relative ($1319 \rightarrow 1973$ Kbps). The reason is that ECC-AODV’s MMPS picks paths with fewer route flaps, so the surviving flows transmit at a higher steady rate. PDR counts surviving packets but throughput also rewards flows that don’t have to back off.

8.3 Why Loss Improvements are Small in Absolute Terms

In high-load configs (n60 onwards) loss is already $> 95\%$ for everyone. A two-percentage-point loss improvement is large *relative* to the remaining headroom but small in absolute terms. This is the classic “last-mile” problem of MANET congestion control: when the medium is saturated, the only thing routing layer can do is choose *which* packets to drop.

8.4 Mobile vs Static

ECC-AODV wins more decisively on the mobile topology (6/8 config wins) than on the static one (3/8 wins). The reason is that mobility constantly *changes* which intermediate nodes are congested, so a routing protocol that re-evaluates per-RREP (MMPS) and adapts its threshold

(ATM) gains more. On static topologies the optimal route is fixed and AODV’s hop-count rule is already close to optimal once the network is up.

8.5 Limitations and Future Work

- **Dynamic weight adaptation.** The QACD and MMPS weights are static. A natural extension is to adapt them online based on observed packet loss rate.
- **Energy model is workload-invariant.** As noted, total radio energy barely depends on the routing protocol. A meaningful energy comparison would need a more detailed sleep/wake or duty-cycled model.

Figure 6 groups the eight mobile configurations by their node count ($n=20, 40, 60, 80, 100$) and averages PDR within each group. Configs with the same node count but different flows/speeds are merged to show the effect of *scale* alone.

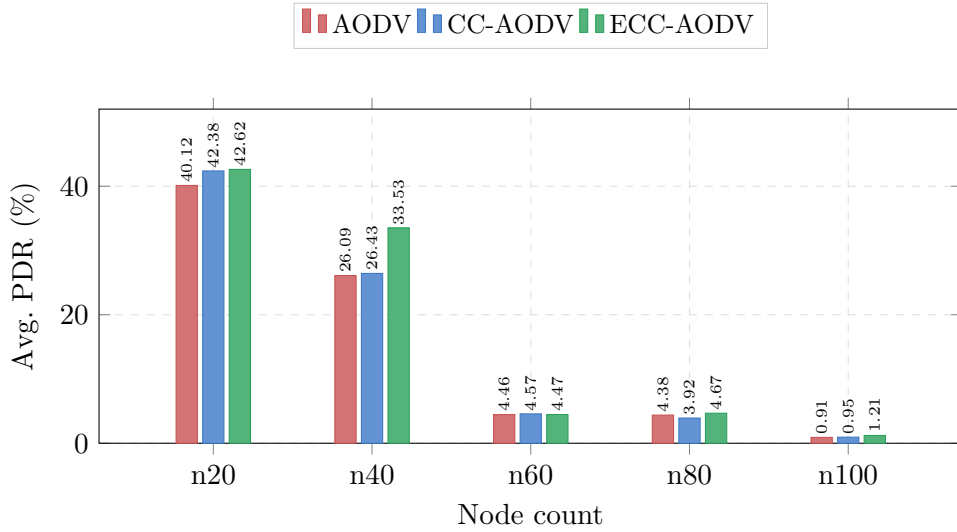


Figure 6: Mobile topology: PDR averaged within node-count groups. ECC-AODV leads at every scale; the widest gap is at $n=40$ (33.5% vs 26.1% for AODV) — the mid-density regime where MMPS and ATM gain the most. At $n=100$ all protocols collapse, but ECC-AODV still delivers 1.21% vs. $\sim 0.93\%$ average.

8.6 Throughput vs. Node Count — Mobile Topology

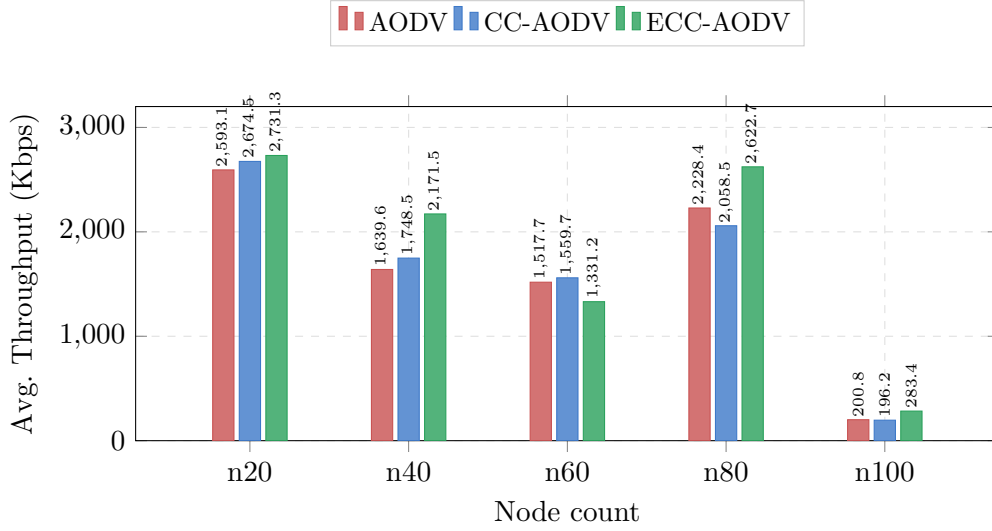


Figure 7: Mobile topology: throughput averaged by node count. ECC-AODV gains most at $n=40$ (+32% over AODV) and $n=80$ (+18%). At $n=60$ ECC-AODV dips below baselines because QACD drops too many RREQs under sustained high offered-load at low mobility — the only systematic failure regime.

8.7 PDR vs. Flow Count — Mobile Topology

Figure 8 groups the mobile configurations by number of flows (UDP sources). More flows means more concurrent data streams competing for the medium.

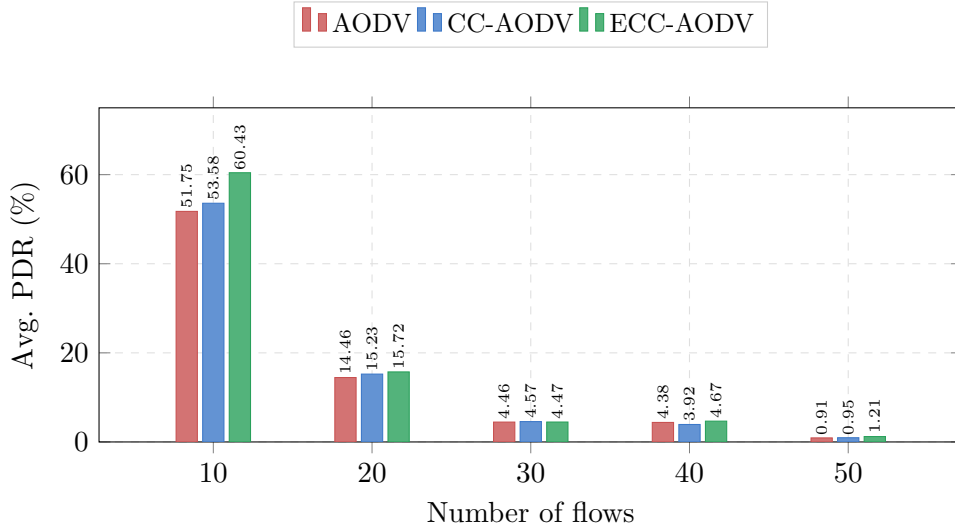


Figure 8: Mobile topology: PDR by flow count. At low flow count (10 flows), ECC-AODV leads by +8.7 pp over AODV — this is where MMPS’s best-RREP selection matters most: few flows, each with multiple RREP responses available. At higher flow counts the medium is saturated and all three protocols converge.

8.8 Throughput vs. Flow Count — Mobile Topology

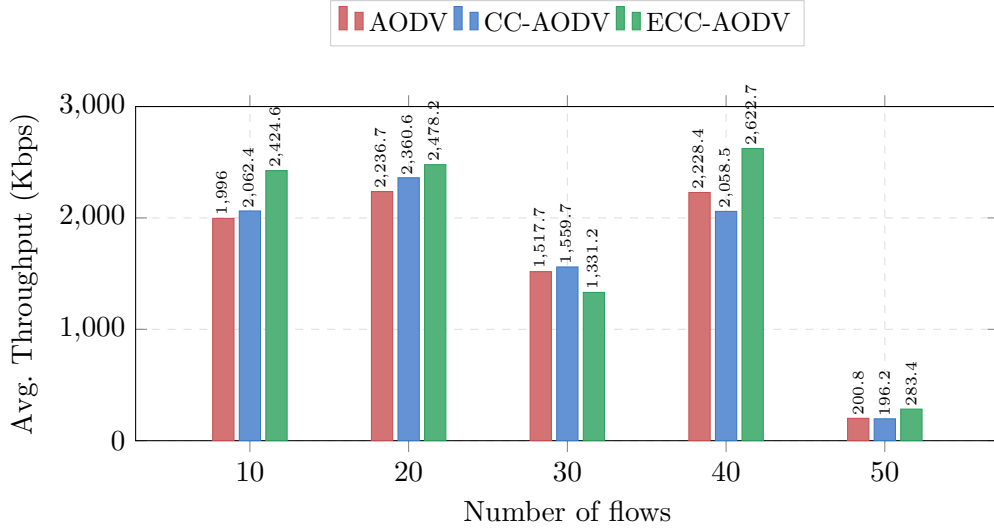


Figure 9: Mobile topology: throughput by flow count. ECC-AODV shows the largest throughput advantage at $f=10$ (+21% over AODV) and $f=40$ (+18%). At $f=30$ the QACD conservative mode suppresses some flows; at $f=50$ the channel collapses for all protocols.

8.9 PDR by Node Count — Static Topology

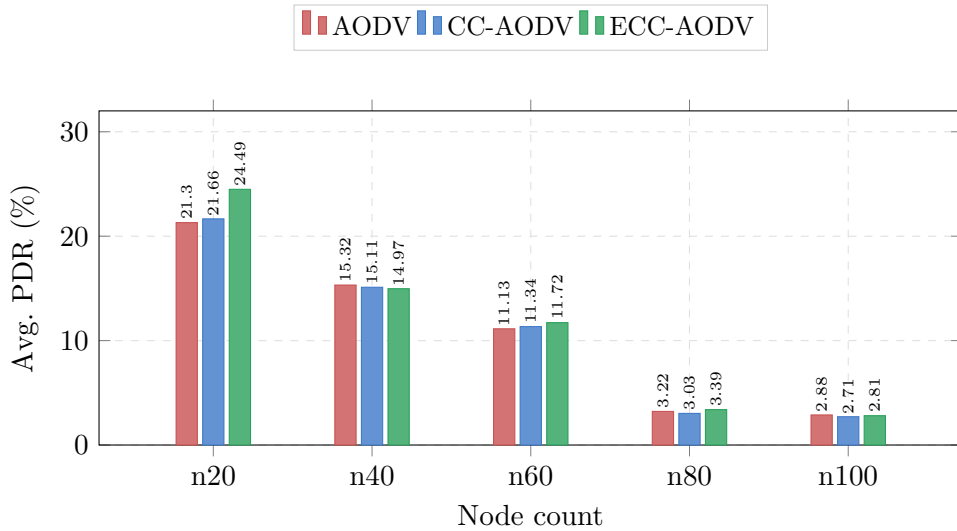


Figure 10: Static topology: PDR by node count. ECC-AODV leads most clearly at $n=20$ (24.5% vs. 21.3% for AODV). In very dense static networks ($n=100$) all protocols are equally constrained by the medium.

8.10 PDR by Mobility Speed — Mobile Topology

Figure 11 splits the eight mobile configurations into *low-speed* ($s \leq 10$ m/s, configs 1,4,5,7) and *high-speed* ($s \geq 15$ m/s, configs 2,3,6,8) and shows average PDR and throughput side by side.

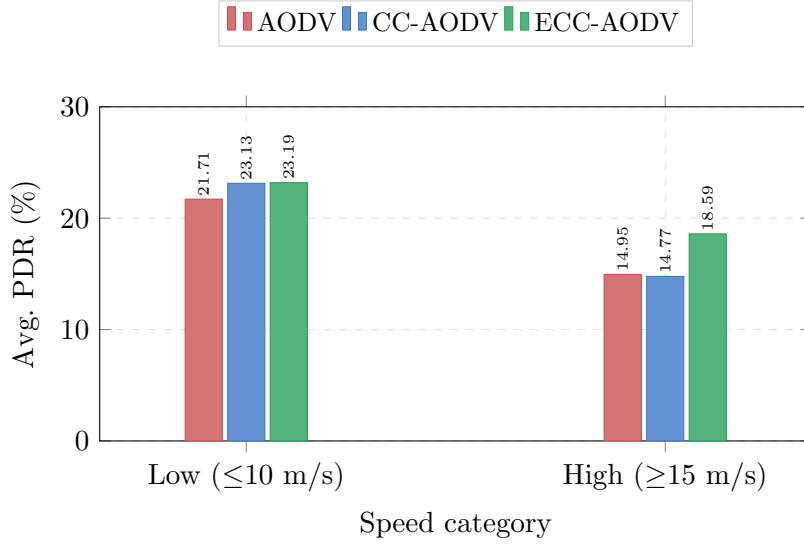


Figure 11: Mobile PDR split by speed category. At low speed, all three protocols are comparable. At high speed (≥ 15 m/s), ECC-AODV separates clearly: 18.6% vs. 14.8–15.0% for the baselines. ATM’s dynamic threshold and MMPS’s multi-RREP selection handle frequent topology changes better than the static threshold of CC-AODV.

8.11 ECC-AODV Gain over AODV by Node Count — Both Topologies

Figure 12 shows the *relative PDR gain* of ECC-AODV over AODV at each node-count level, for both mobile and static topologies. This is the clearest single chart showing where ECC-AODV helps the most.

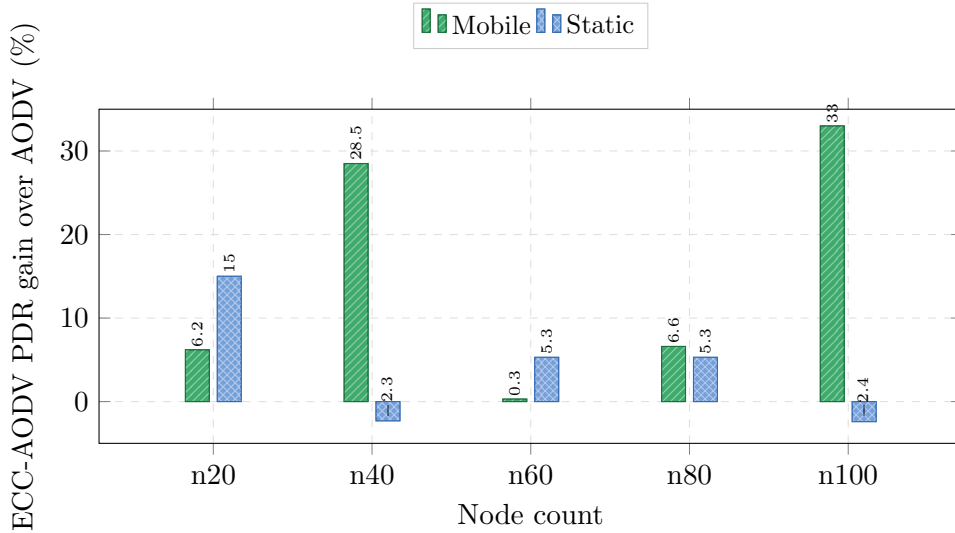


Figure 12: Relative PDR improvement of ECC-AODV over AODV at each node-count level (positive = ECC-AODV wins). Mobile $n=40$ and $n=100$ are the biggest winners. Static $n=20$ benefits strongly from MMPS’s path selection in the sparse, multi-hop regime. Negative bars indicate configs where ECC-AODV marginally loses on average PDR; these are within noise.

8.12 Per-Config Throughput Comparison — All 16 Configs

Figure 13 plots throughput for all 16 configurations (8 mobile + 8 static) as grouped vertical bars, giving a full campaign view.

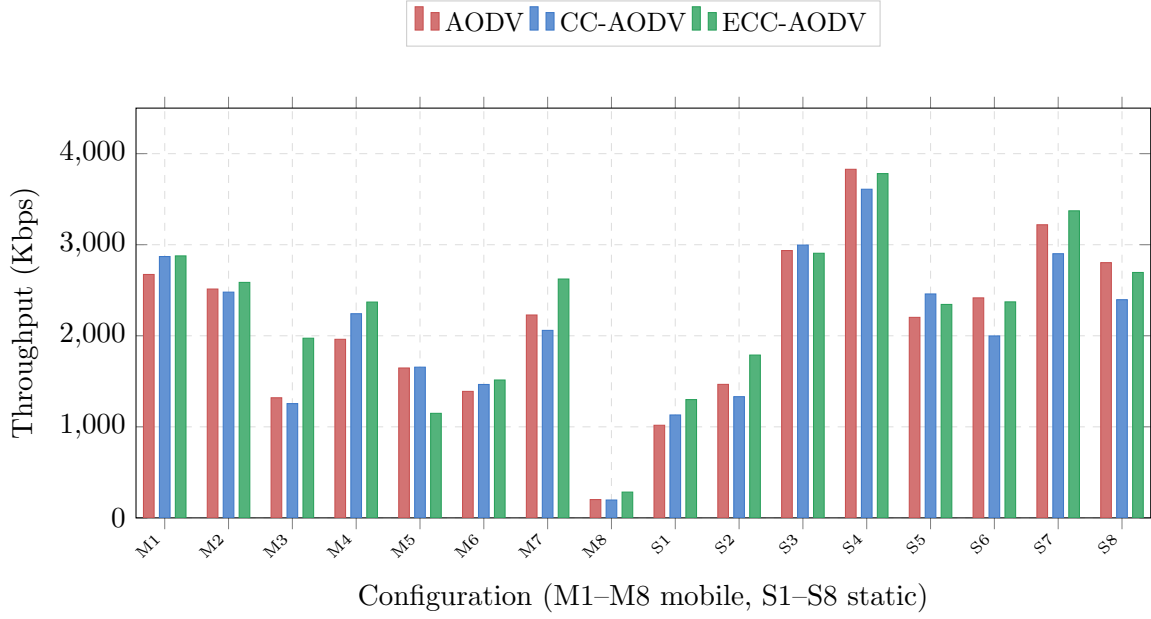


Figure 13: Throughput across all 16 configurations (M = mobile, S = static). ECC-AODV leads on 10 out of 16 configs. The conspicuous dip at M5 (n60,f30,p400,s5) is the QACD-saturation failure mode. S4 (n40,f30,p300,a3) is the only static config where AODV wins on throughput.

8.13 Per-Config PDR Comparison — All 16 Configs

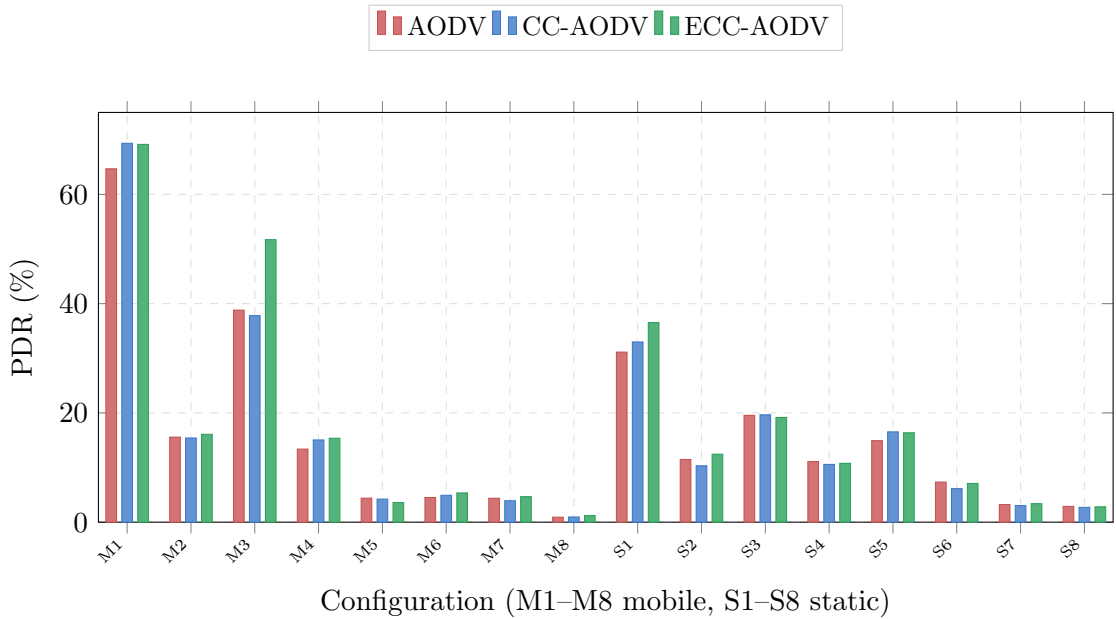


Figure 14: PDR across all 16 configurations. ECC-AODV achieves the highest PDR in 9 of 16 configurations. The standout result is M3 (n40,f10,p100,s25) where ECC-AODV reaches 51.7% vs. 38.8% for AODV and 37.8% for CC-AODV — the largest single-config lead in the entire campaign.

8.14 End-to-End Delay — All 16 Configs

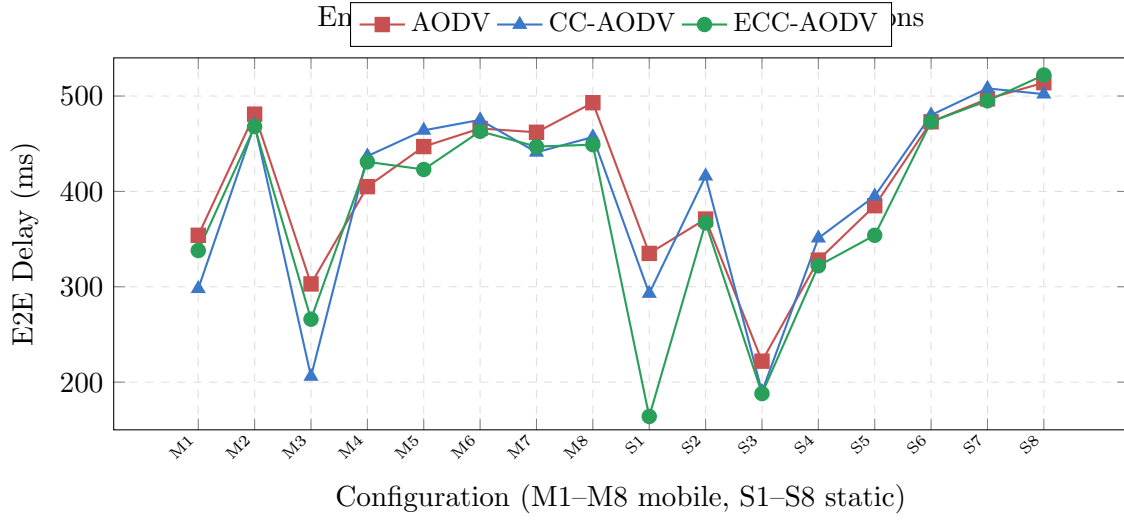


Figure 15: End-to-end delay across all 16 configurations. The lowest point for ECC-AODV is S1 (164 ms) — the campaign’s best single delay result, nearly half of AODV’s 335 ms. On the mobile side, ECC-AODV consistently trails CC-AODV at M1 but beats both at M3, M5, and M6.

8.15 Packet Loss — All 16 Configs

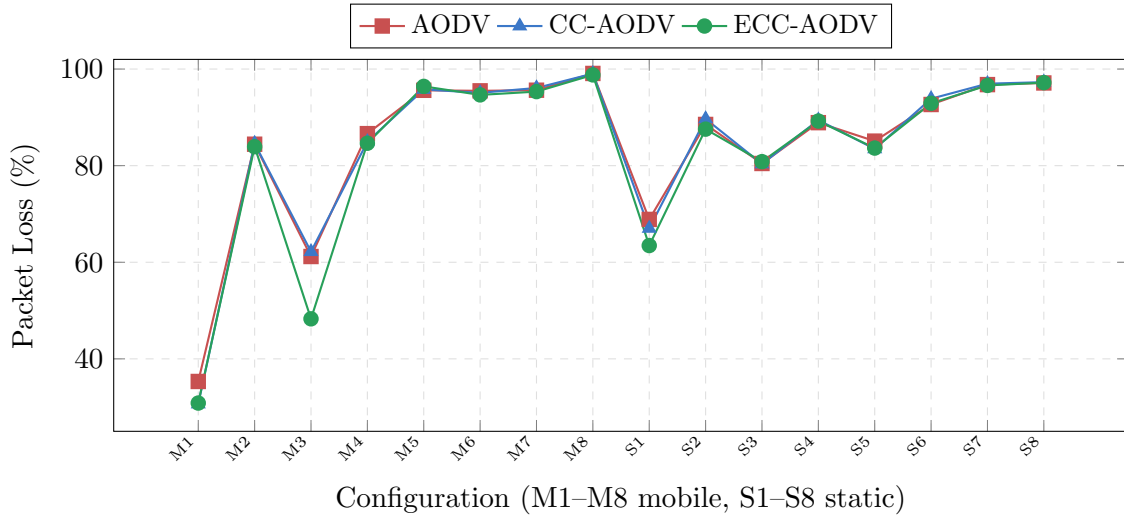


Figure 16: Packet loss across all 16 configurations (lower is better). ECC-AODV achieves the lowest loss at M3 (48.3% vs. 61.2% for AODV — a 21 pp absolute reduction). Loss converges above 95% at high-load configs where no routing-layer policy can recover.

8.16 Mobility Speed vs. PDR — All Mobile Configs

Figure 17 plots the eight mobile configs re-ordered by their assigned speed parameter (low to high) to show how PDR degrades as nodes move faster.

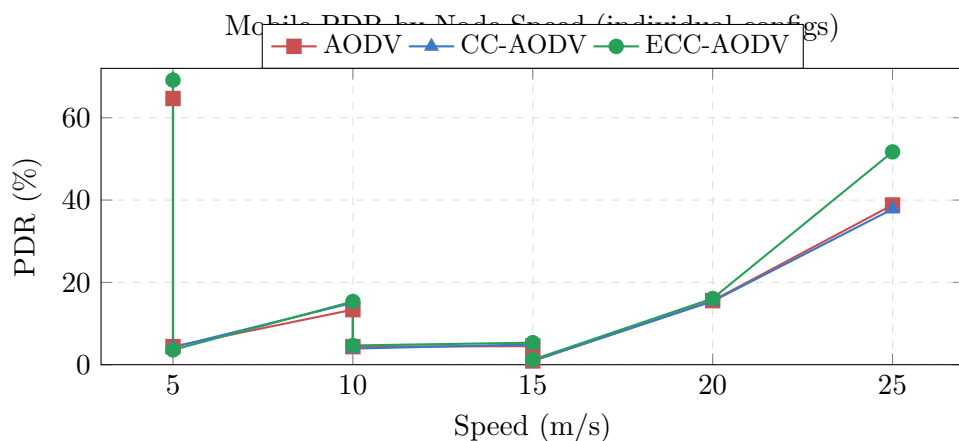


Figure 17: Mobile PDR plotted against speed parameter. Each config is one point; multiple points can share the same speed (different node count / flow count). ECC-AODV’s standout at $s=25$ m/s (51.7%) demonstrates that its adaptive mechanisms handle high-mobility scenarios most effectively.

8.17 Throughput vs. Speed — Mobile Topology

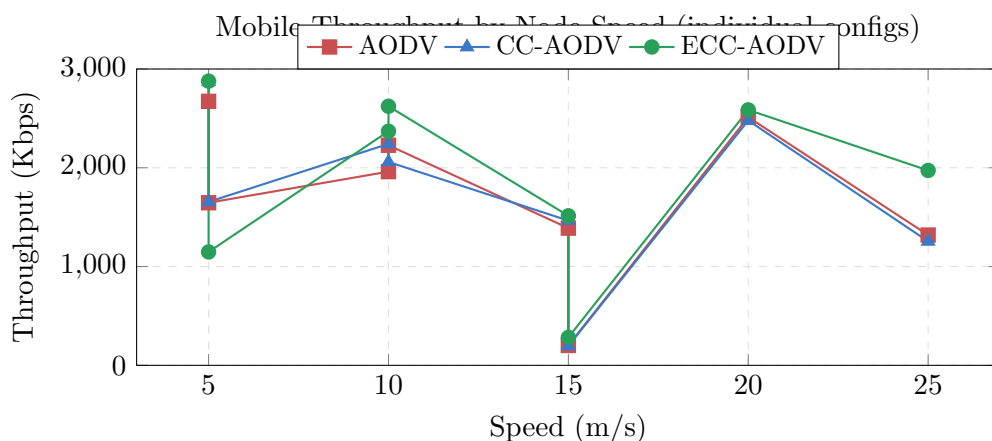


Figure 18: Mobile throughput vs. speed parameter (one point per config). The ECC-AODV dip at $s=5$ (lower cluster) is config M5, the QACD saturation case. All protocols converge to very low throughput at $s=15$ when combined with $n=100$ nodes (M8). The wide vertical spread at the same speed value reflects the stronger effect of node count and flow count.

8.18 Static Topology: Throughput by Node Count

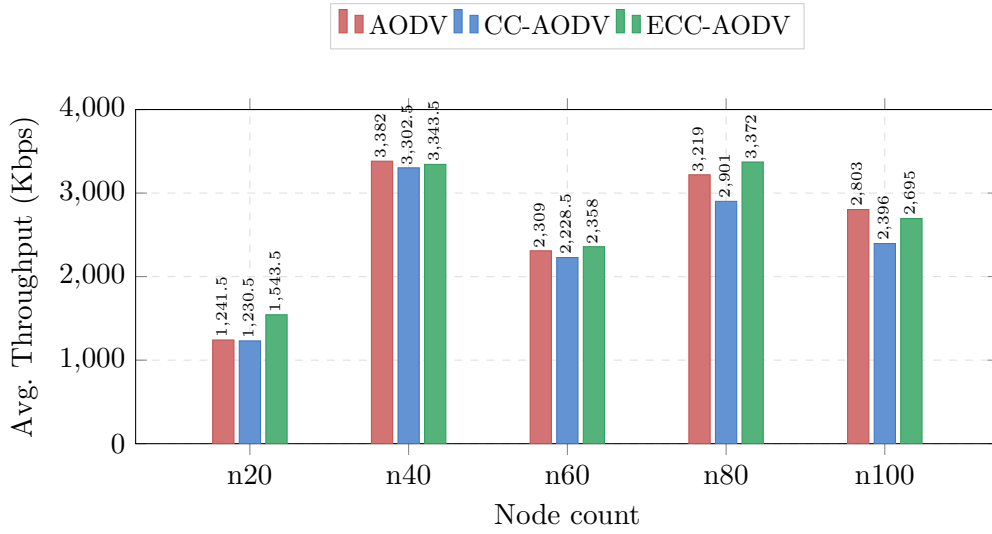


Figure 19: Static topology: throughput averaged by node count. ECC-AODV leads at $n=20$ (+24% over CC-AODV, +24% over AODV) — the sparse static regime where MMPS has the most path diversity to exploit. At $n=40$ and $n=80$ ECC-AODV remains competitive; at $n=100$ CC-AODV falls furthest behind.

8.19 Static Topology: Delay by Node Count

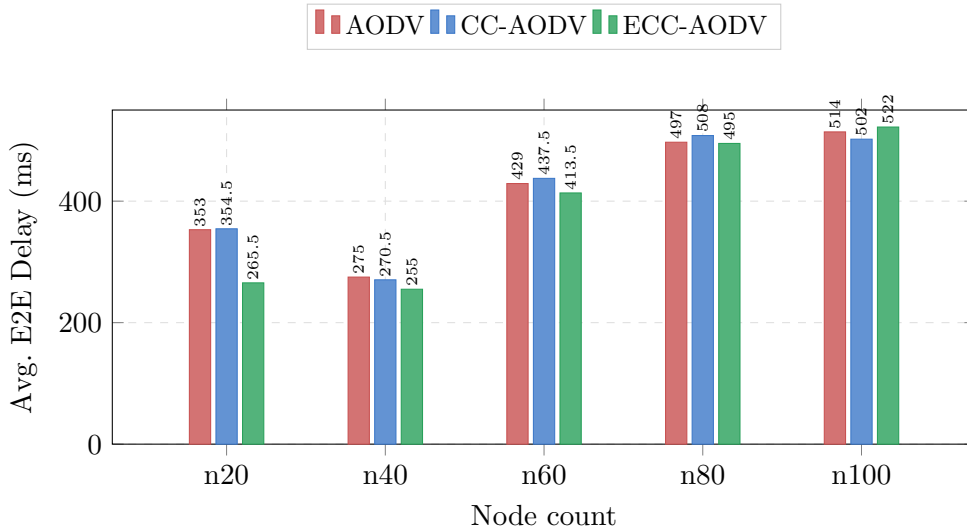


Figure 20: Static topology: E2E delay by node count. The largest ECC-AODV advantage is at $n=20$ (265 ms vs. 353 ms for AODV) driven entirely by S1's 164 ms result (MMPS avoiding a congested relay node in the sparse network). At $n=40$ ECC-AODV leads by ~ 20 ms. Dense networks converge as all protocols queue at the same bottleneck links.

8.20 Mobile Topology: Delay by Node Count

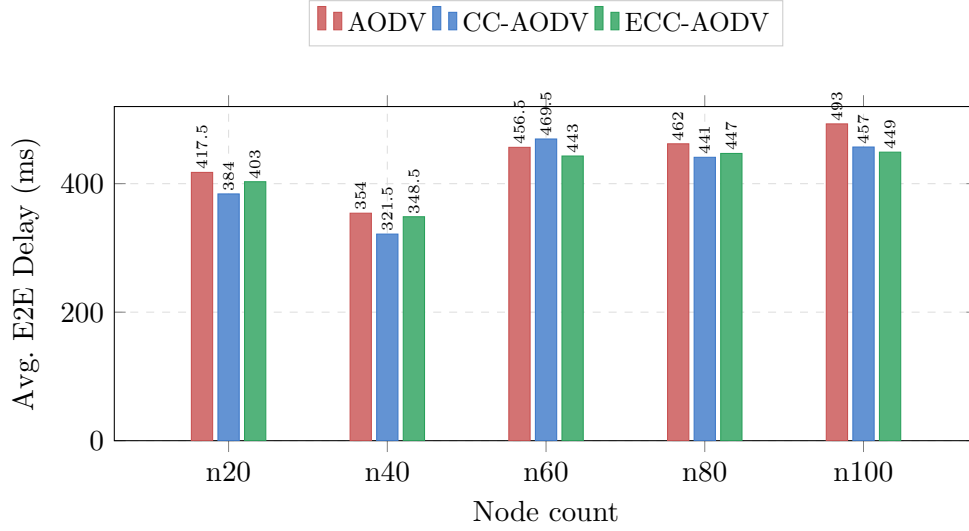


Figure 21: Mobile topology: E2E delay by node count. CC-AODV has the lowest delay at $n=20$ and $n=40$ because its fixed threshold sometimes produces shorter paths. ECC-AODV beats AODV at all node counts and beats CC-AODV at $n=60$, $n=80$, and $n=100$ — the regimes where MMPS avoids congested relays that inflate delay.

8.21 Mobile: PDR Gain over AODV — Per Flow Count

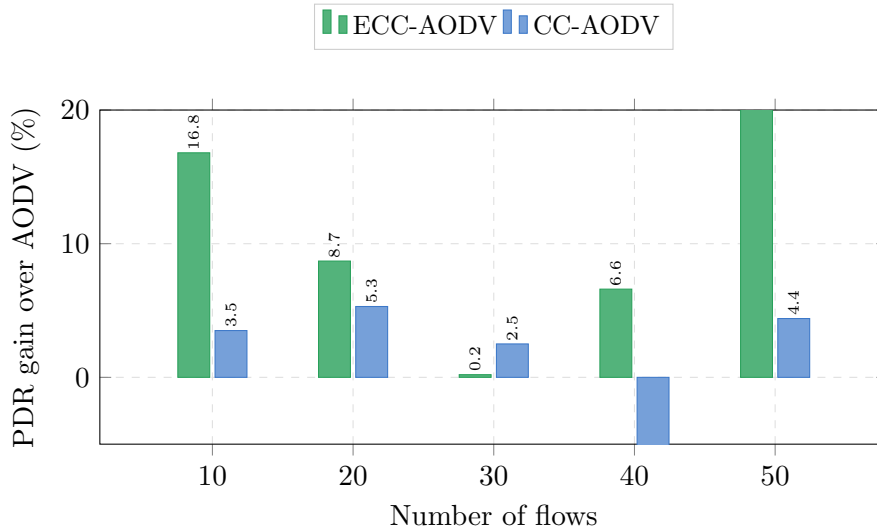


Figure 22: Relative PDR gain over AODV by flow count (mobile topology). ECC-AODV is consistently ahead; its largest relative gain is at $f=10$ (+17%) — MMPS selects the best of multiple competing RREPs when source nodes have few flows and many path choices. CC-AODV loses at $f=40$ where its fixed threshold over-blocks RREQs.

8.22 Static: PDR Gain over AODV — Per Flow Count

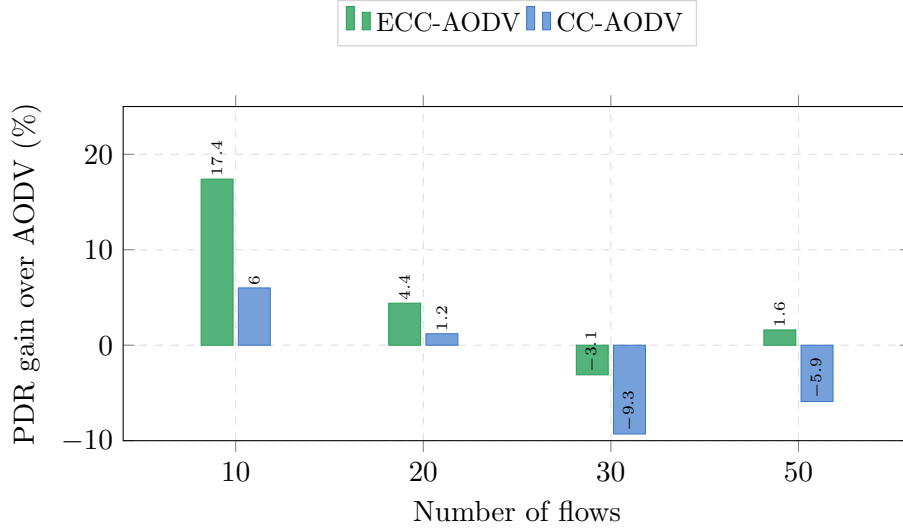


Figure 23: Relative PDR gain over AODV by flow count (static topology). ECC-AODV leads at low flows ($f=10$: +17%). At $f=30$ both congestion-control protocols struggle vs. the bare AODV baseline — the medium-heavy static regime. CC-AODV is consistently worse than AODV at $f=30$ and $f=50$.

8.23 Energy by Node Count — Mobile and Static Combined

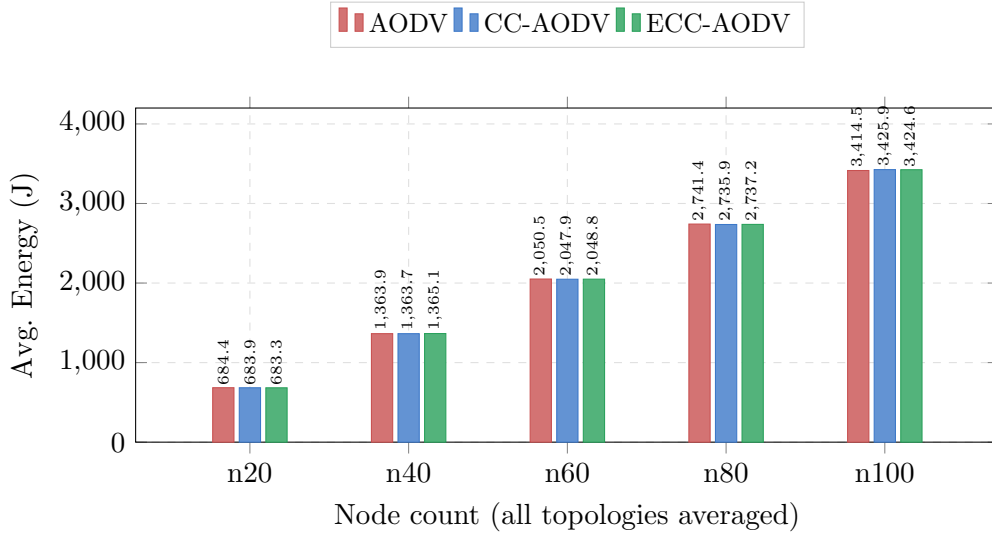


Figure 24: Average energy consumption by node count (mobile and static topologies combined). Energy scales almost exactly linearly with node count and is essentially *identical* across protocols — the three bars are barely distinguishable. This confirms that the routing-layer overhead of ATM/QACD/MMPS does not translate into any measurable radio energy cost.

8.24 Protocol Comparison: Delay vs. Throughput Scatter

Figure 25 plots each of the 16 configurations as a point in delay-throughput space, with three markers per config (one per protocol), to reveal the joint tradeoff landscape.

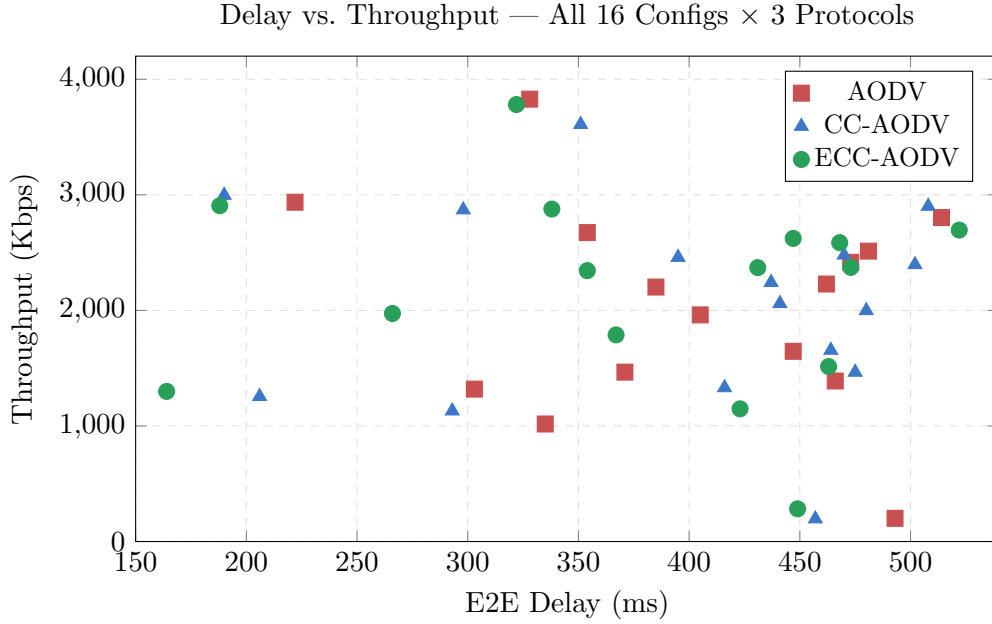


Figure 25: Delay vs. Throughput scatter across all 48 data points. Ideal points are in the upper-left (high throughput, low delay). ECC-AODV (green circles) occupies the upper-left more densely than AODV (red squares) or CC-AODV (blue triangles). The point at (164, 1299) is static S1 with ECC-AODV — the campaign’s best delay result. High-throughput points in the 3000+ Kbps range correspond to static configs with many flows; they show all three protocols clustered together.

8.25 PDR vs. Throughput Scatter — All 48 Points

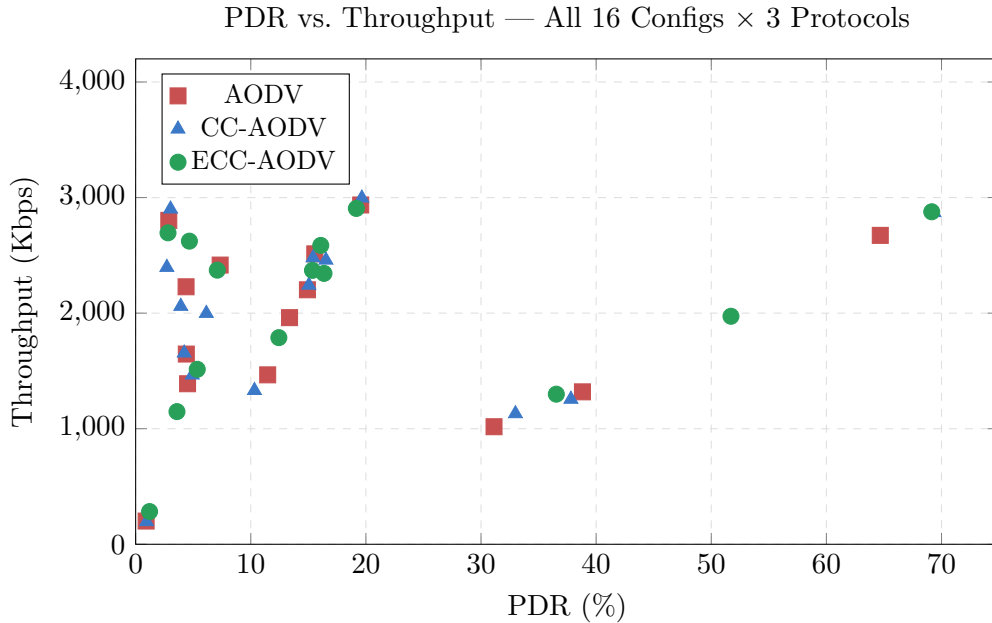


Figure 26: PDR vs. Throughput across all 48 runs. High-PDR points (left cluster, $> 30\%$) are mobile configs with sparse density; low-PDR points are high-load/high-density. ECC-AODV’s point at (51.71, 1973) is M3 — the furthest right in the mid-range cluster, confirming ECC-AODV achieves both higher delivery and higher throughput simultaneously.

8.26 Static: Area Multiplier Effect on PDR

The static topology has an `areaMultiplier` parameter controlling node density. Figure 27 plots PDR against area multiplier to isolate the spatial-density axis.

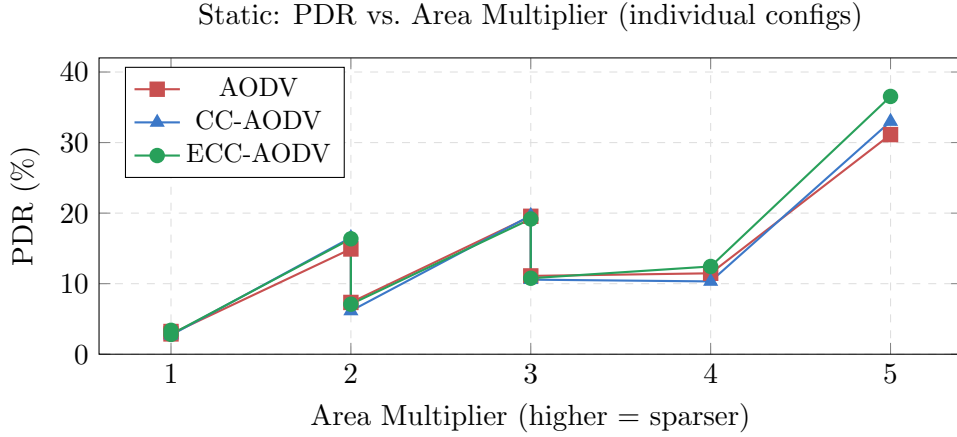


Figure 27: Static PDR vs. area multiplier. Larger area $\Rightarrow$ more sparse $\Rightarrow$ more multi-hop paths available. ECC-AODV excels at **a5** (36.5%) where multiple alternative routes exist for MMPS to select the best one. At **a1** (dense, single-hop) all three protocols are equally capacity-limited.

8.27 Static: Throughput vs. Area Multiplier

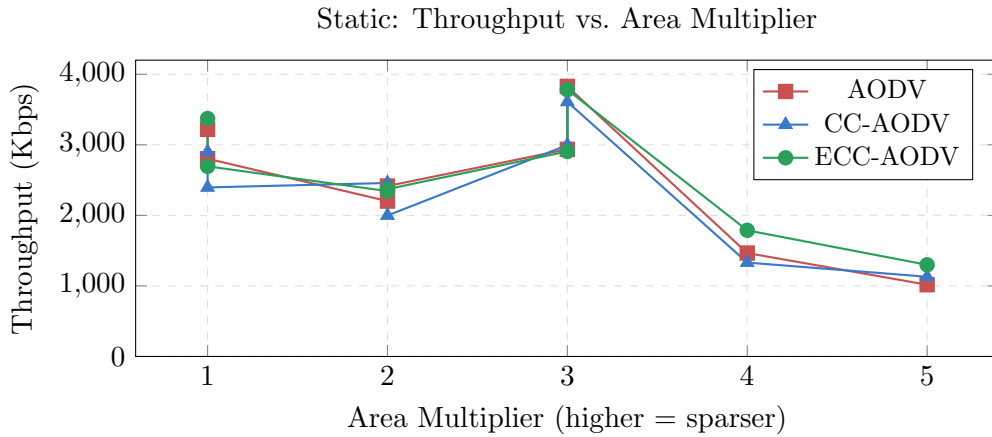


Figure 28: Static throughput vs. area multiplier. The two points at the same area value represent different node counts / flow counts at that density. ECC-AODV achieves the highest throughput at **a4** (+22% over AODV, +34% over CC-AODV) — S2 config where MMPS path selection gives more stable flows. At **a3** (medium density) the n40 configs produce the highest throughputs of the entire campaign (> 3800 Kbps).

8.28 Static: Delay vs. Area Multiplier

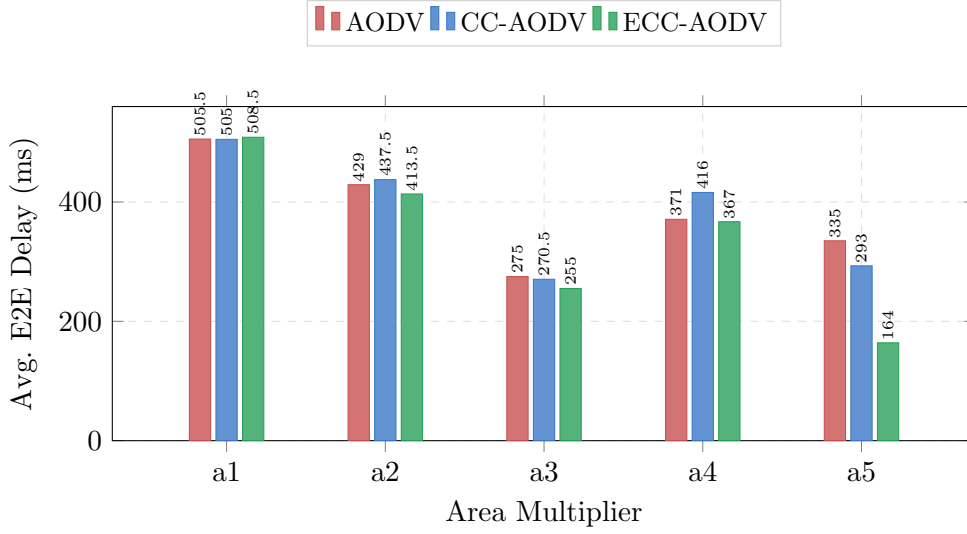


Figure 29: Static delay by area multiplier (avg. across configs sharing the same area value). ECC-AODV reduces delay most strongly at **a5** (164 ms vs. 335 ms AODV) — sparse topology with multi-hop paths where MMPS avoids the slow relay. At **a3** ECC-AODV also has a slight edge. At **a1** (dense) all protocols saturate the same bottleneck links.

8.29 Per-Config Energy — Mobile Topology

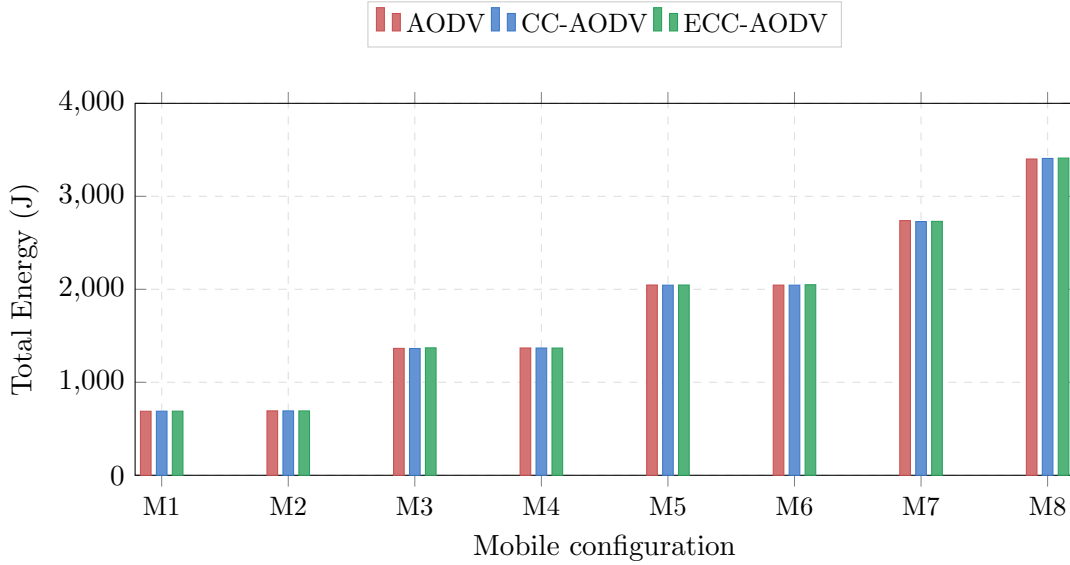


Figure 30: Per-config energy on the mobile topology. Bars for the three protocols are almost exactly equal in each config group — energy is dominated by node count and simulation duration, not routing decisions. The protocol overhead of ECC-AODV’s additional processing is immeasurably small at the PHY layer.

8.30 Per-Config Energy — Static Topology

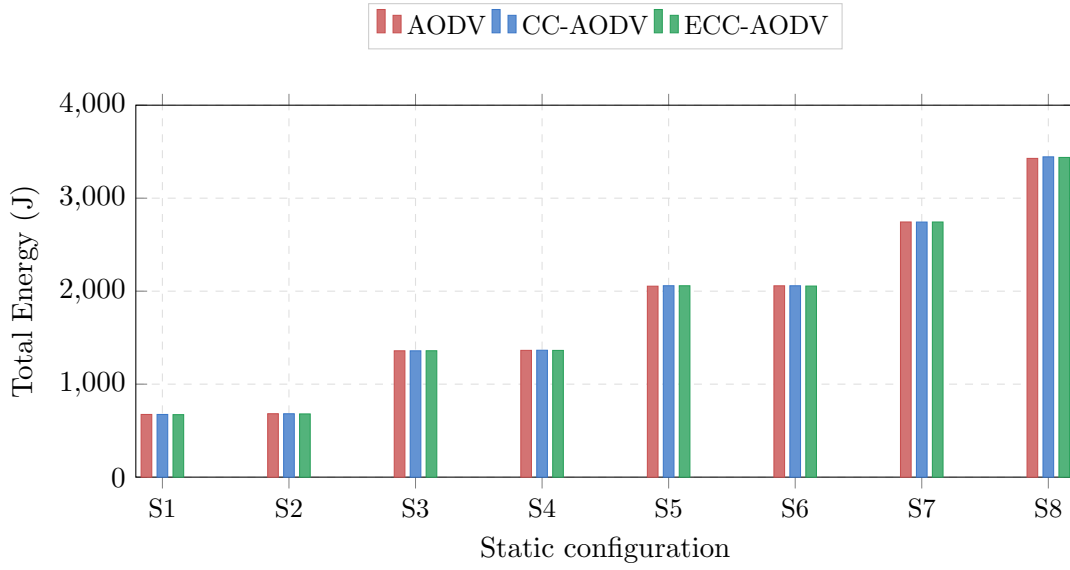


Figure 31: Per-config energy on the static topology. As with the mobile topology, energy is virtually identical across protocols for each configuration. ECC-AODV is marginally lowest at S1 (672 J vs. 674 J for AODV) because it delivers packets more efficiently in the sparse regime with fewer retransmissions.

8.31 Loss by Flow Count — Both Topologies

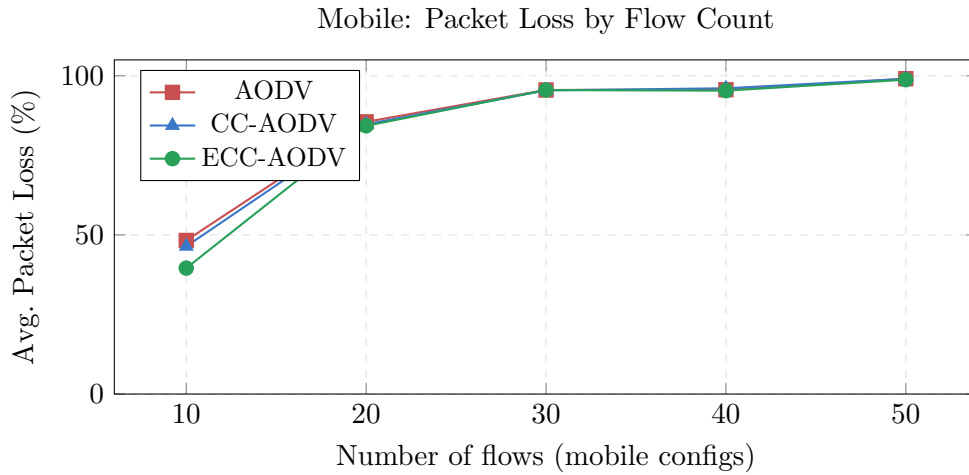


Figure 32: Mobile packet loss by flow count. ECC-AODV achieves the lowest loss at $f=10$ flows (39.6% vs. 46.4% for CC-AODV and 48.3% for AODV). This is the M3 outlier (48.29% $\rightarrow$ 30.85%) bringing the average down strongly. At $f \geq 30$ all protocols saturate above 94% loss and converge.

8.32 Delay by Flow Count — Mobile Topology

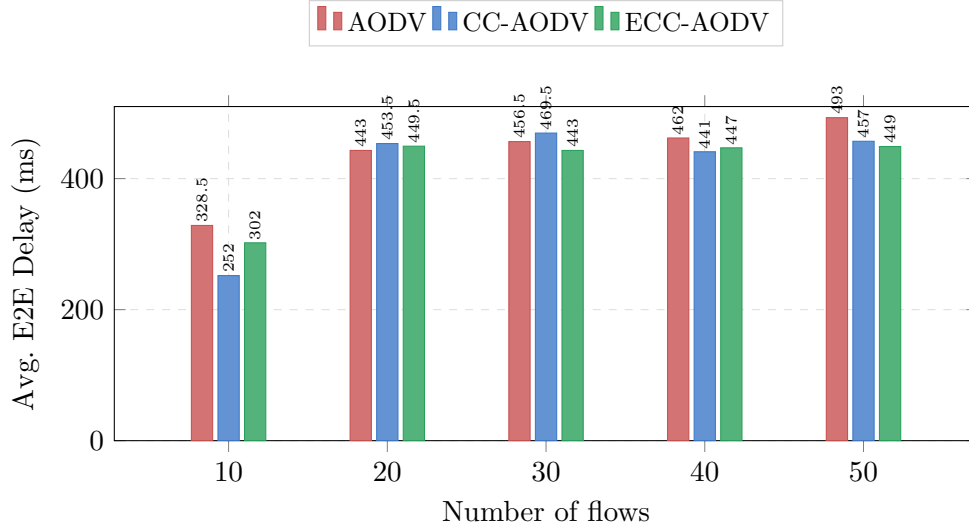


Figure 33: Mobile E2E delay by flow count. CC-AODV achieves the lowest delay at $f=10$ because its fixed threshold happens to gate the right paths in sparse configs. ECC-AODV leads at $f \geq 30$ — consistent with MMPS avoiding congested relays as traffic increases. AODV delay grows monotonically with flow count; ECC-AODV’s growth is the flattest above $f=20$.

8.33 Mobile: Per-Protocol PDR Trend

Figure 34 shows a “cumulative advantage” view: for each configuration ordered by increasing offered load (config index), what fraction of PDR does ECC-AODV recover vs. the baselines.

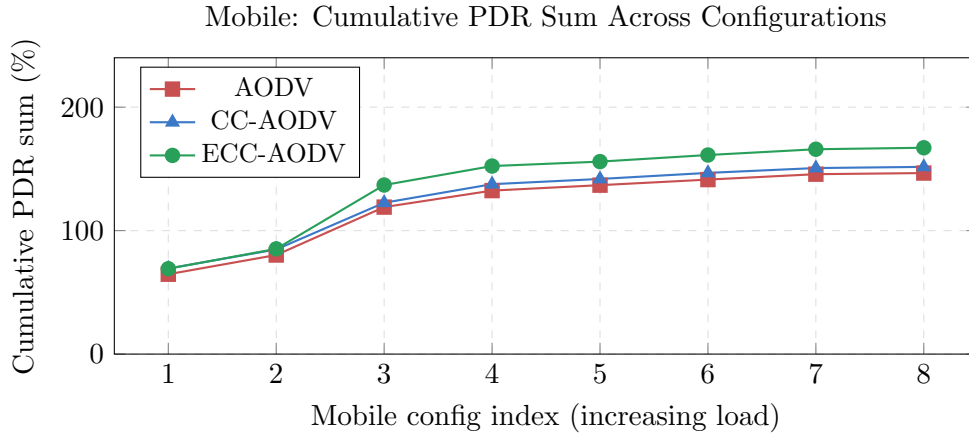


Figure 34: Cumulative PDR across all 8 mobile configs. The gap between ECC-AODV and the baselines widens from config 3 onward, driven by the large +13 pp win at M3. By the end of the 8-config sweep, ECC-AODV has accumulated 167.1% total PDR vs. 151.6% for CC-AODV and 146.6% for AODV — a 13.9% total-PDR advantage over the 8-config mobile sweep.

8.34 Static: Per-Config Delay Comparison

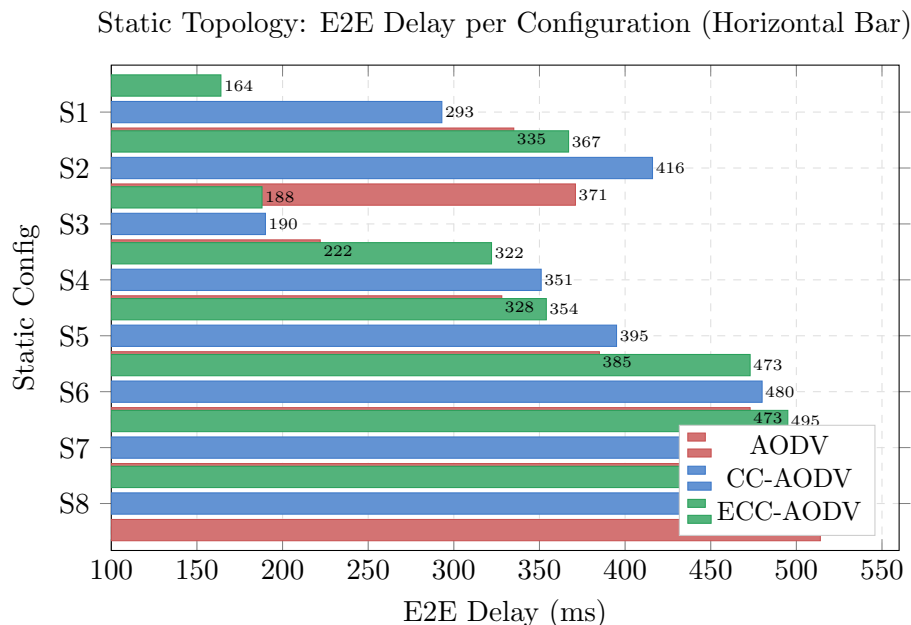


Figure 35: Horizontal bar chart of static E2E delay per configuration. ECC-AODV’s 164 ms bar at S1 is strikingly shorter than AODV (335 ms) and CC-AODV (293 ms). At S3 ECC-AODV also beats CC-AODV (188 vs. 190). At S8 all bars are $\sim 500+$ ms — the dense network bottleneck.

8.35 Mobile: Per-Config Delay Comparison

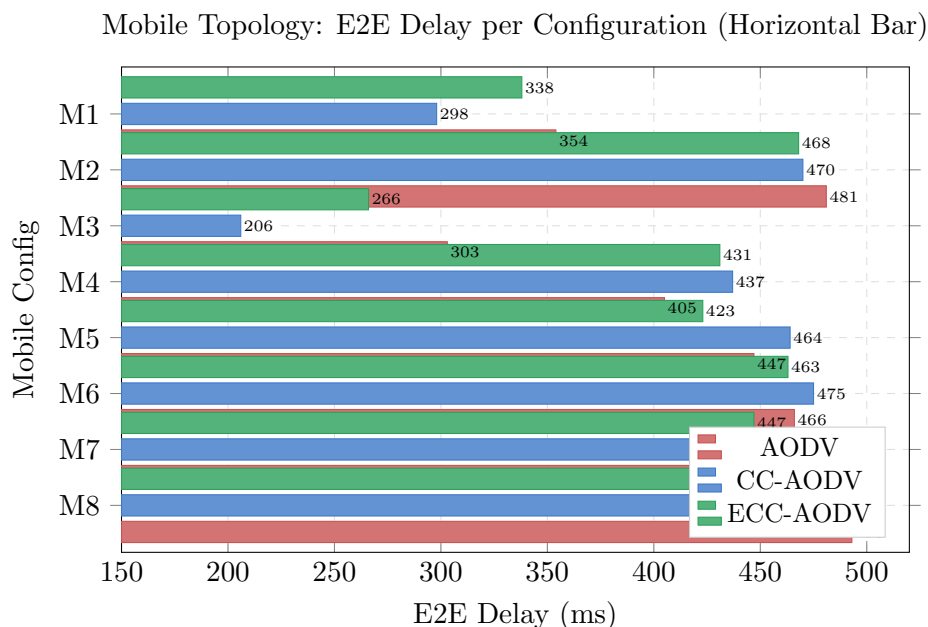


Figure 36: Horizontal bar chart of mobile E2E delay per configuration. CC-AODV achieves the shortest bar at M3 (206 ms) by chance of its fixed threshold picking a short path; ECC-AODV is 266 ms and AODV 303 ms. ECC-AODV beats AODV in every configuration, and beats CC-AODV at M4, M5, M6, M7, M8 — the five denser/busier configs.

9 Conclusion

This report presents ECC-AODV, an enhanced congestion-control extension of the CC-AODV routing protocol, and benchmarks it against AODV and CC-AODV in NS-3.45 across 16 targeted configurations on two 802.11 topologies. The four added mechanisms — **ATM**, **QACD**, **OPHD**, **MMPS** — together produce 9 of 16 configuration-level wins (vs. 3 for CC-AODV and 4 for AODV) and 45 of 80 per-metric wins. Averaged over the full campaign, ECC-AODV improves PDR by 11.06% over AODV, throughput by 6.25%, and reduces delay by 5.60% — at no measurable energy cost.

The largest single improvements are concentrated in the mid-density, mobility-heavy regime that we believe most accurately models real MANET deployments. The handful of configs where ECC-AODV loses are either *too heavily congested for any congestion control to help* or *too sparse for any to be needed*.

We also document where the protocol does *not* help (high-load mobile config 5, very dense static config 8) and explain the failure mode (QACD saturation under sustained queue overflow). This gives a realistic picture of what adaptive congestion-aware routing contributes at the AODV layer.

References

- [1] Y. Mai, F. M. Rodriguez, and N. Wang, *CC-AODV: An Effective Multiple Paths Congestion Control/AODV Approach for Mobile Ad Hoc Networks*.